Renal System: Pharmacology

DIURETICS

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Session Objectives

- Discuss the basic principles of diuretic therapy
- Apply physiological reabsorption mechanisms in the proximal convoluted tubule (PCT) to discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics
- Apply physiological reabsorption mechanisms in the Loop of Henle (LOH) to discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of loop diuretics
- Apply physiological reabsorption mechanisms in the distal convoluted tubule (DCT) to discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of thiazides
- Apply physiological reabsorption mechanisms in the cortical collecting tubule (CCT) to discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of potassium-sparing diuretics
- Explain the rationale behind combination therapy with loop diuretics and thiazides
- Review the physiology of vasopressin (ADH, AVP) release and actions at V1 and V2 receptors
- Discuss the indications, pharmacodynamics, pharmacokinetics, and ADRs of vasopressin and desmopressin
- Discuss the indications, pharmacodynamics, pharmacokinetics, and ADRs of vasopressin antagonists
- Provide examples of pharmacokinetic and pharmacodynamic drug interactions with vasopressin receptor ligands

Section 1: Basics of Diuretic Therapy and Agents Acting in the Proximal Tubule

- Objectives:
  - Discuss the basic principles of diuretic therapy
  - Apply physiological reabsorption mechanisms in the proximal convoluted tubule (PCT) to discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics

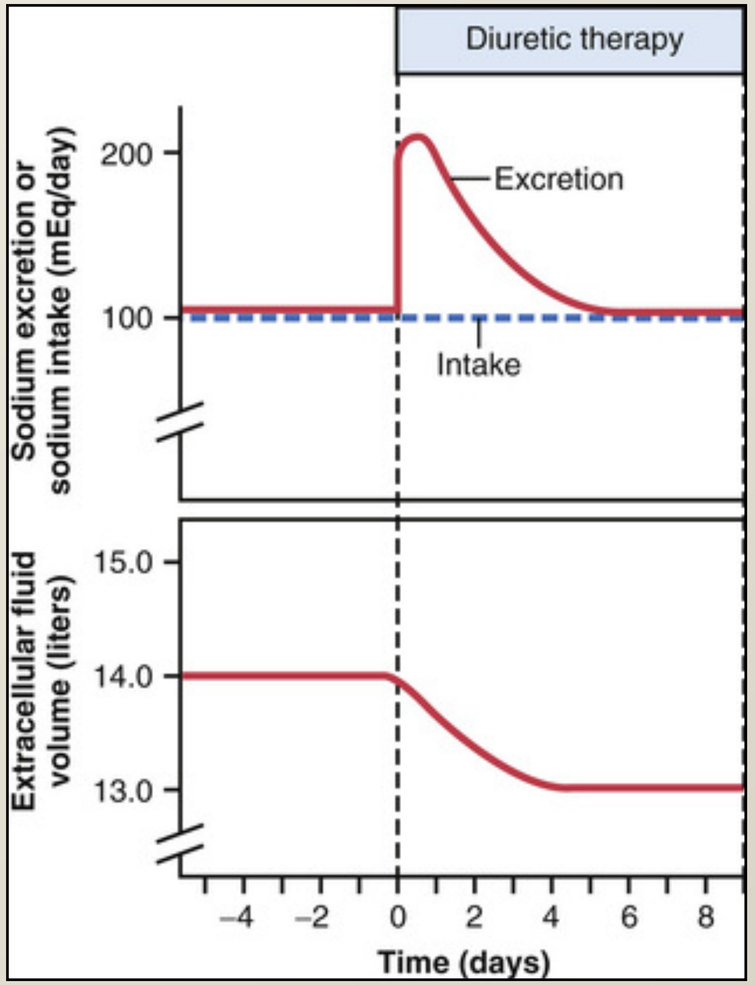
Common Indications of Diuretic Therapy

- Diuretic agents treat **volume overload states** by increasing urine volume

- **Edematous states**
  - Heart failure
  - Kidney Disease and Renal Failure
  - Hepatic Cirrhosis
  - Idiopathic edema

- **Non-edematous states**
  - Hypertension
  - Diabetes insipidus
  - Renal & Cardiac protection
  - Glaucoma
  - Acute head trauma
  - Nephrolithiasis*
  - Hypercalcemia*

- Mostly used in **combination** with other medications and changes to diet:
  - Ex. ACE inhibitors, angiotensin receptor blockers, ß-antagonists, calcium channel blockers, etc.



Guyton and Hall Textbook of Medical Physiology, Figure 32-1

***Not necessary a volume overload state**

Major Classes of Diuretic Agents

- Diuretics modify either *salt* or *water* excretion
  - ***Natriuretic agents*** increase urine volume by excretion of Na^+:
    - Natriuretic agents **concentrate** Na^+ in the tubule by blocking reabsorption into blood
    - Water follows solute, resulting in urine volume increase
  - ***Aquaretic agents*** increase urine volume by excretion of solute-free water
    - *Aquaretic agents do not increase the excretion of Na^+*
- Therapeutic effectiveness of diuretic agents varies:
  - Duration of action/dosage frequency
  - Dietary salt intake
  - **Distribution to nephron tubule and rate of filtrate movement**
  - **Site of action in the nephron tubule**

Natriuretic Agents	Aquaretic Agents
Loop diuretics (e.g., furosemide)	**Osmotic diuretics** (e.g., mannitol)
Thiazide and thiazide-like diuretics (e.g., chlorothiazide)	**ADH antagonists** (e.g., lithium, vaptan agents)
Potassium-sparing diuretics (e.g., amiloride, spironolactone)	—
Carbonic anhydrase inhibitors (e.g., acetazolamide)	—

The Vast Majority of Diuretic Agents are Natriuretic

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  - **Site of action in the nephron tubule**

Natruiretic →

Aquaretic →

THIAZIDE DIURETICS

Chlorothiazide DIURIL
Chlorthalidone GENERIC ONLY
Hydrochlorothiazide (HCTZ) MICROZIDE
Indapamide GENERIC ONLY
Metolazone ZAROXOLYN

LOOP DIURETICS

Bumetanide BUMEX
Ethacrynic acid EDECRIN
Furosemide LASIX
Torsemide GENERIC ONLY

POTASSIUM-SPARING DIURETICS

Amiloride MIDAMOR
Eplerenone INSPRA
Spironolactone ALDACTONE
Triamterene DYRENIUM

CARBONIC ANHYDRASE INHIBITORS

Acetazolamide DIAMOX

OSMOTIC DIURETICS

Mannitol OSMITROL

Lippincott Illustrated Reviews: Pharmacology 8e

Most diuretic agents act from within the nephron tubule

- Diuretics enter nephron tubule by one of two routes:
  - **Filtration** through glomerulus
  - **Secretion** into proximal tubule through organic acid ion transporters
    - Interactions occur with other substances that use the organic acid ion transport system

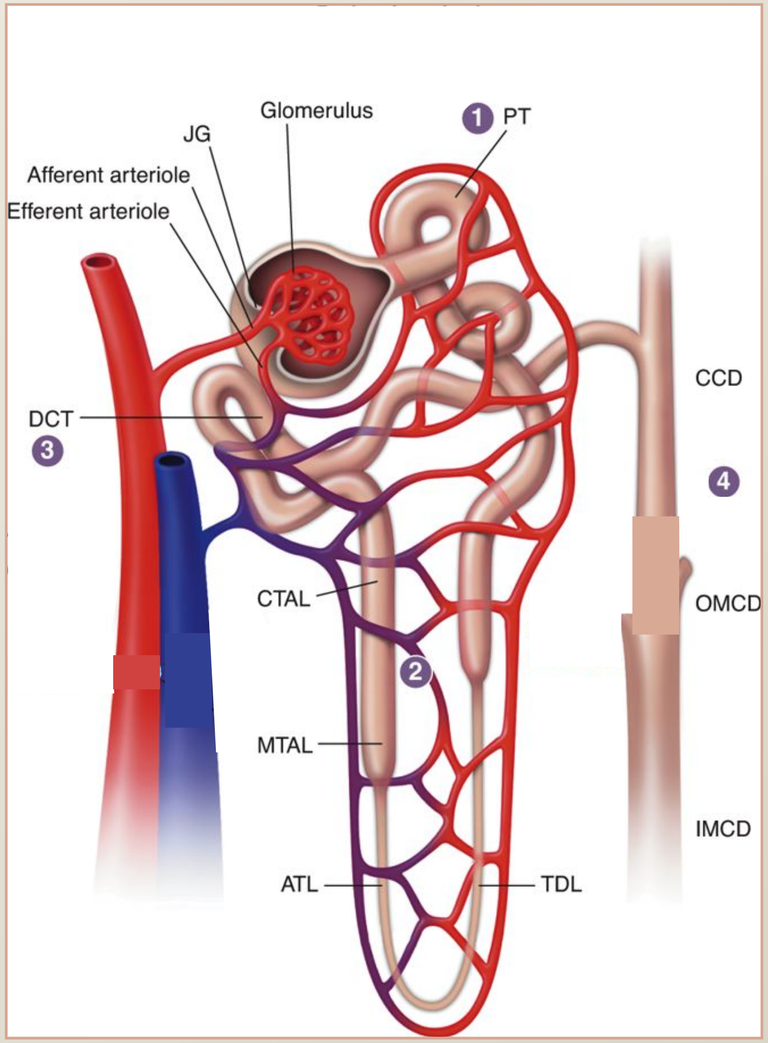


Fig. 21-5, *Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy*

Diuretic effects depend on the **site** of action in the nephron tubule

Diuretic Agents	Site of Action
Carbonic anhydrase inhibitors (e.g., acetazolamide)	Proximal Tubule
Osmotic diuretics (e.g., mannitol)	Proximal Tubule; Loop of Henle (descending limb)
Loop diuretics (e.g., furosemide)	Loop of Henle (ascending limb)
Thiazide and thiazide-like diuretics (e.g., chlorothiazide)	Distal Convoluted Tubule
Potassium-sparing diuretics (e.g., amiloride, spironolactone)	Collecting Tubule/Duct
ADH antagonists (e.g., lithium, vaptan agents)	Collecting Duct

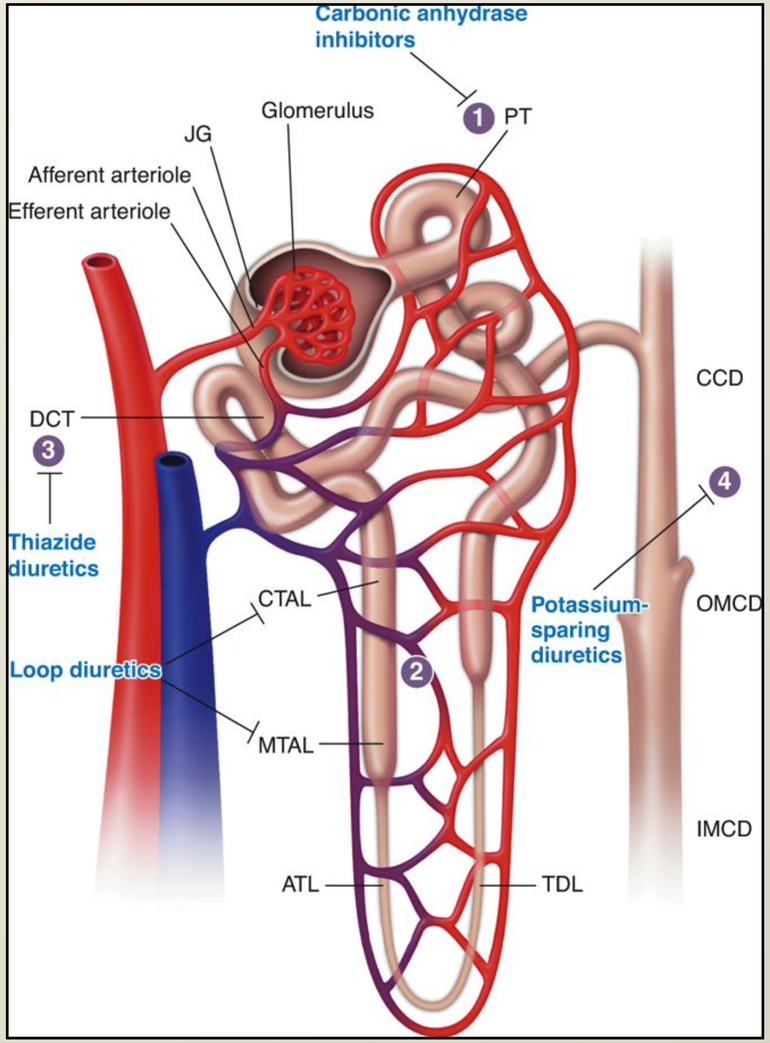
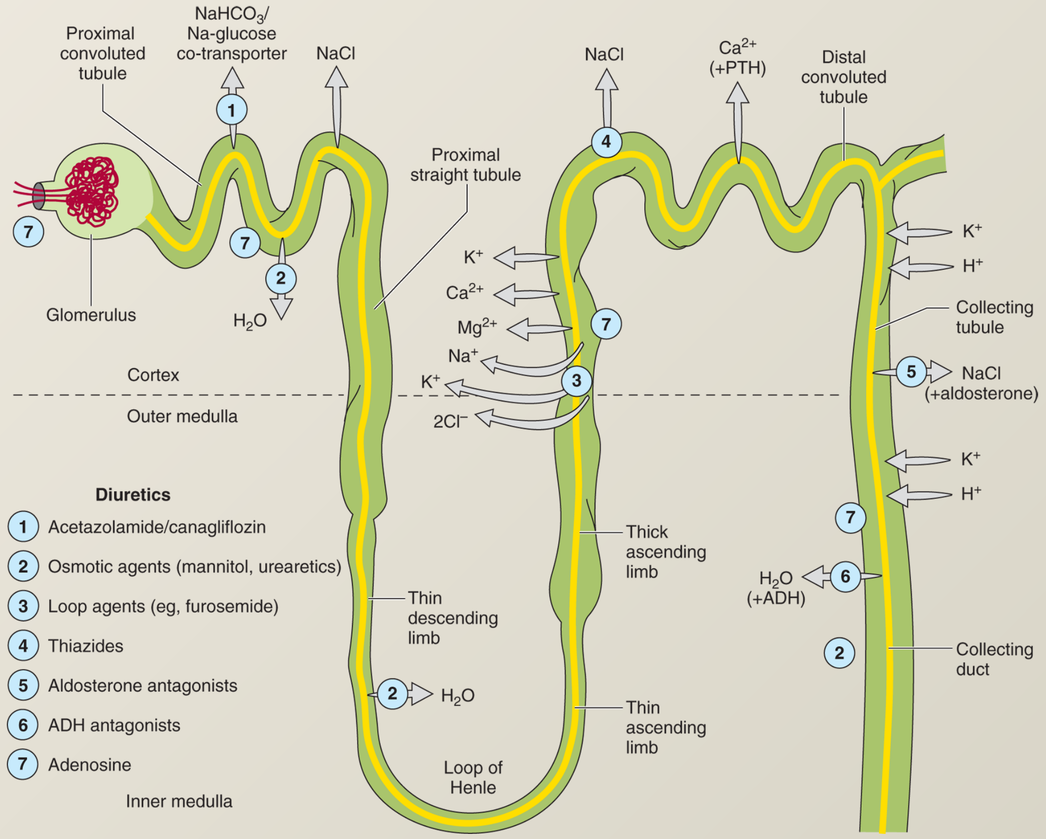


Fig. 21-5, *Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy*

Diuretic effects depend on the site of action in the nephron tubule

Segment	Functions	Water Permability	Primary Transporters and Drug targets	Diuretic with major action
Proximal convoluted tubule (PCT)	Reabsorption of 65% of filtered Na^+/K^+/Ca^{2+} and Mg^{2+}; 85% of $NaHCO_3$, ~100% of glucose and amino acids	Very high	Na/H (NHE3), carbonic anhydrase; Na/glucose cotransporter 2 (SGLT2)	Carbonic anhydrase inhibitors, Osmotic Diuretics
Proximal tubule, straight segments	Secretion and reabsorption of organic acids and bases, including uric acid and most diuretics	Very high	Acid (eg, uric acid) and base transporters	Osmotic Diuretics
Thin descending limb of Henle's loop	Passive reabsorption of water	High	Aquaporins	Osmotic Diuretics
Thick ascending limb of Henle's loop (TAL)	Active reabsorption of 15–25% of filtered Na^+/K^+/Cl^-; secondary reabsorption of Ca^{2+} and Mg^{2+}	Very low	Na/K/2Cl (NKCC2)	Loop diuretics
Distal convoluted tubule (DCT)	Active reabsorption of 4–8% of filtered Na^+ and Cl^-; Ca^{2+} reabsorption under PTH control	Very low	Na/Cl (NCC)	Thiazides
Cortical collecting tubule (CCT)	Na^+ reabsorption (2–5%) coupled to K^+ and H^+ secretion	Variable (controlled by vasopressin)	Na channels (ENaC), K channels,[1] H^+ transporter,[1] aquaporins	K^+-sparing diuretics
Medullary collecting duct	Water reabsorption under vasopressin control	Variable (controlled by vasopressin)	Aquaporins	Vasopressin antagonists

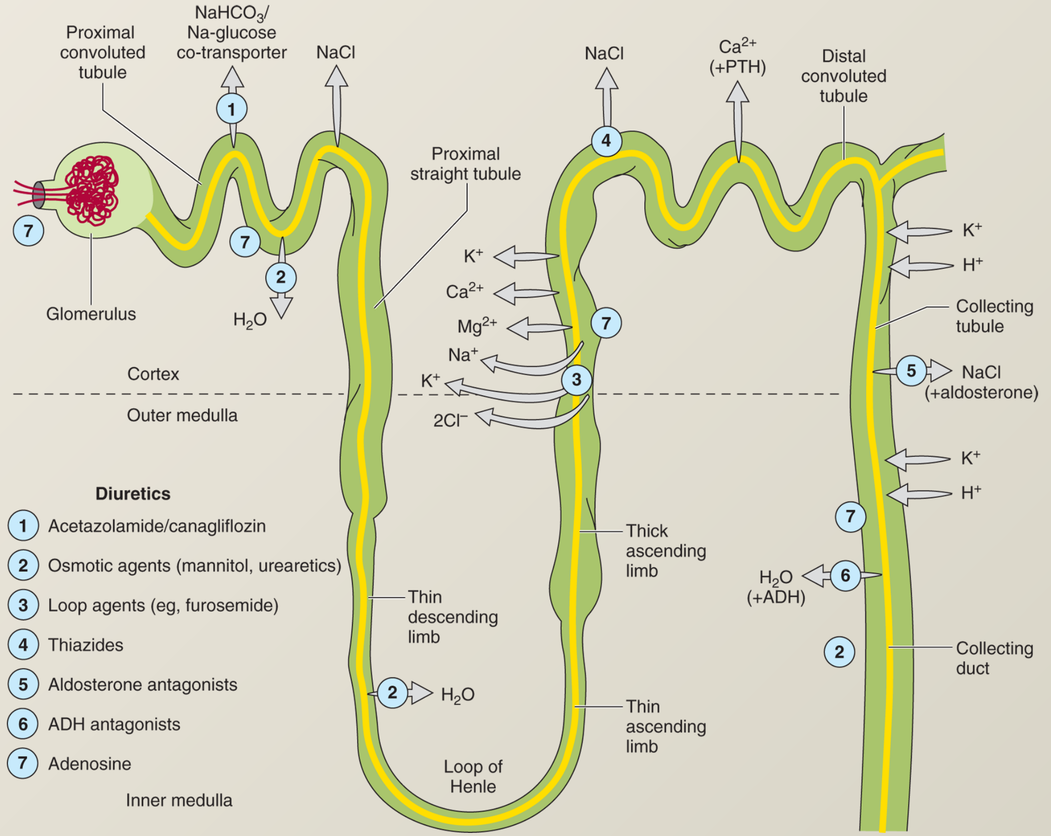


Modified from "Basic and Clinical Pharmacology," 16e.

"Basic and Clinical Pharmacology," 16e.

Diuretic effects depend on the **site** of action in the nephron tubule

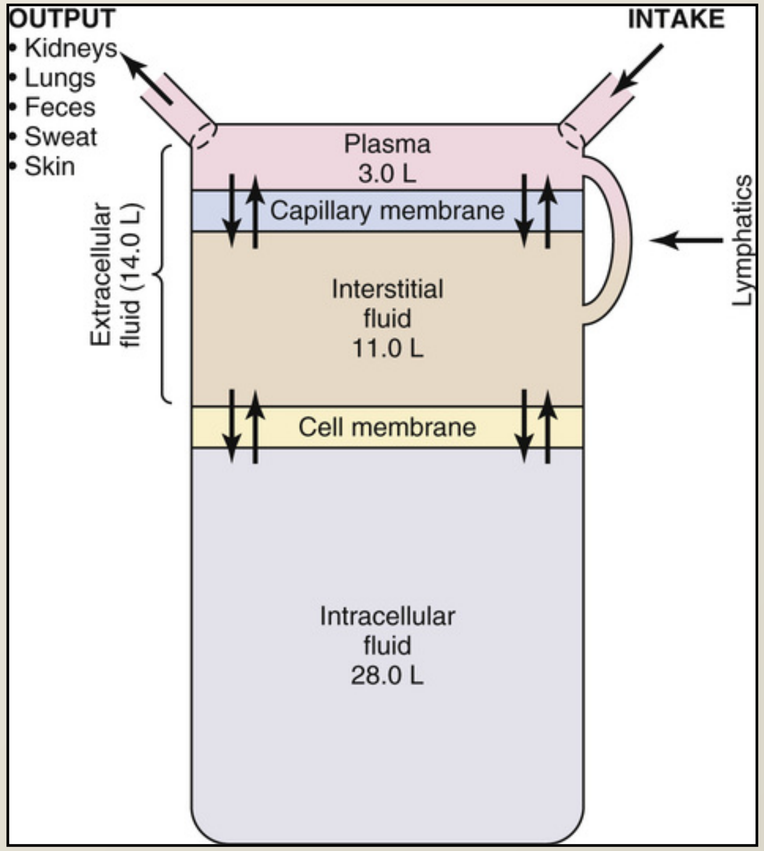
- *Important to understand mechanisms of **ion handling** and the **amount of Na^+ reabsorbed** at each nephron segment*
- Changes in ionic composition in one tubule segment alters ionic composition presented to distal regions
    - (i.e., changes in ionic composition “earlier” in the tubule affect reabsorption “later” in the tubule)
- <u>General ADRs</u>:
    - Electrolyte imbalances
    - ECF loss
    - Off-target ADRs
- Mechanism of diuresis impacts both **indication** and specific **ADRs** of each drug class



“Basic and Clinical Pharmacology,” 16e.

Common Adverse Responses to Diuretic Agents: **Rapid loss of blood volume and ECF**

- Rapid diuretic therapy can result in low blood volume
  - Poor perfusion of organs (i.e., acute kidney injury), hypernatremia, etc.
  - Reduced volume without concomitant reduction in HCO_3^- can result in metabolic alkalosis
- Diuretic therapy may also reduce overall ECF, as reduced plasma volume can “pull” fluid from interstitial spaces
- Usually greatest risk with loop diuretics
  - *Patient monitoring is essential when using powerful diuretics*



Guyton and Hall Textbook of Medical Physiology, Figure 25-1

Common Adverse Responses to Diuretic Agents: **Electrolyte imbalances**

- Most ADRs associated with diuretics are fluid and electrolyte imbalances
  - On-target ADRs can be anticipated and managed
  - Electrolyte changes depend on diuretic duration of action, potency, and site of action

Common Electrolyte ADRs associated with Diuretic Use
Hyponatremia
Acid-Base Changes
Hypokalemia, Hyperkalemia
Hypomagnesemia
Hyperuricemia

National Kidney Foundation Primer

Common Adverse Responses to Diuretic Agents: **Off-target ADRs**

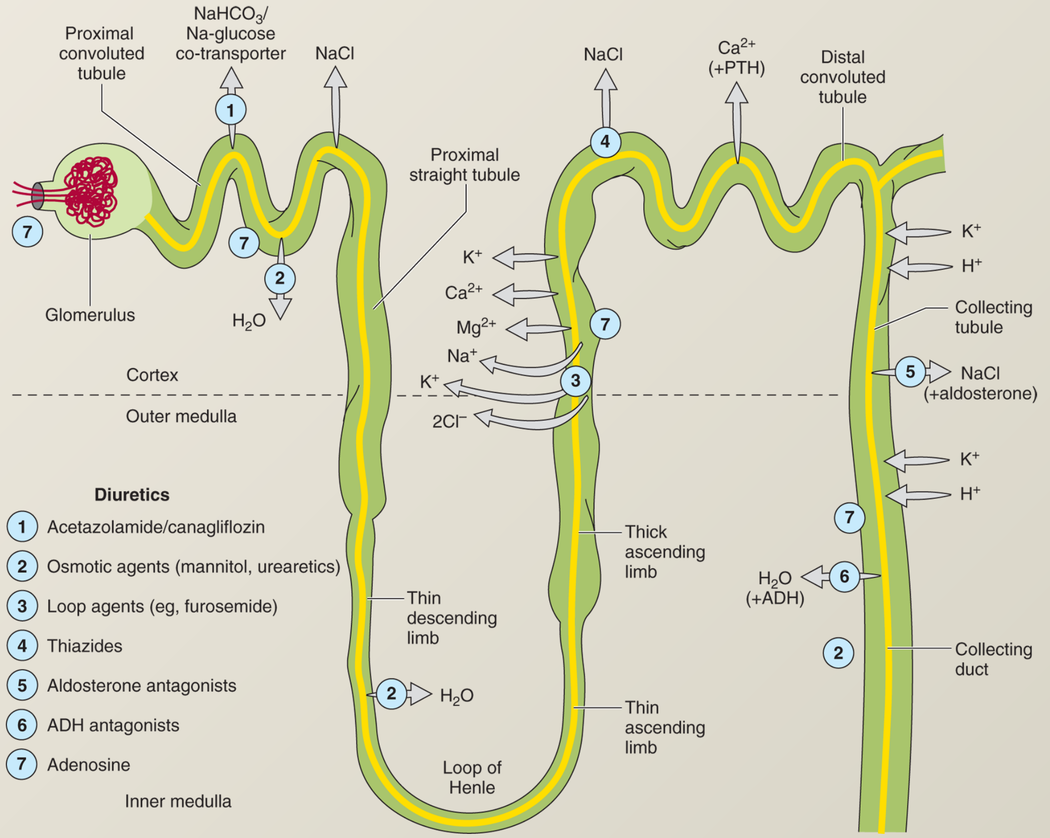
- Off-target ADRs are harder to anticipate in individual patients:
  - **Ototoxicity** (loop diuretics), impotence, etc.
- Allergic reactions may occur with sulfonamide-based drugs
- Diuretic agents have high potential for drug-drug interactions

Common Off-target ADRs associated with Diuretic Use
Hyperlipidemia
Otoxicity
Drug allergy
Drug-drug interactions

National Kidney Foundation Primer

Classes of Diuretic Agents and Locations of Action

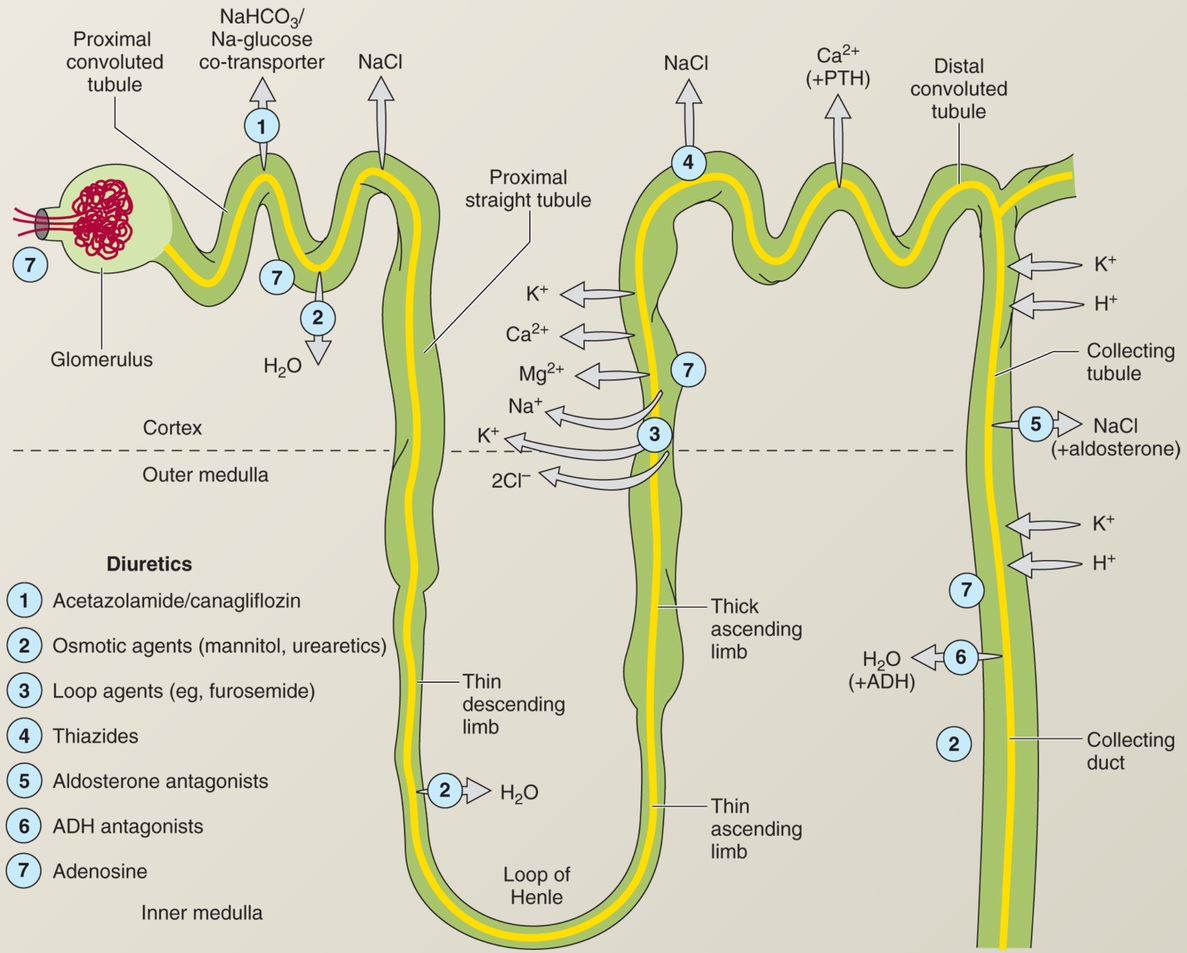
	Diuretic Agents	Site of Action
Hour 1	**Carbonic anhydrase inhibitors**	Proximal Tubule
	Osmotic diuretics	Proximal Tubule; Loop of Henle (descending limb)
	Loop diuretics	Loop of Henle (ascending limb)
Hour 2	**Thiazide and thiazide-like diuretics**	Distal Convoluted Tubule
	Potassium-sparing diuretics	Collecting Tubule/Duct
Hour 3	**ADH antagonists**	Collecting Duct



"Basic and Clinical Pharmacology," 16e.

Carbonic Anhydrase Inhibitors

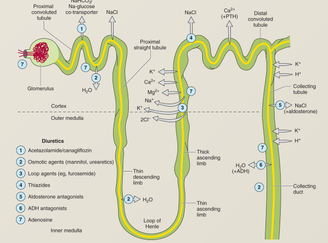
- **Acetazolamide** (Diamox)
- Site of action: **Proximal convoluted tubule**
- **Weak diuretic** agents due to reabsorption mechanisms later in the tubule
- **Indications:**
  - Adjunct diuretic agent
  - Treatment of metabolic alkalosis in edematous patients
  - Metabolic alkalosis in altitude sickness
  - Glaucoma
- PO or IV, 100% bioavailable, renally eliminated, $t_{1/2}$ = 6-9 hours
  - Secreted into the proximal tubule



"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the proximal convoluted tubule (PCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics

Normal reabsorption mechanisms in the PCT



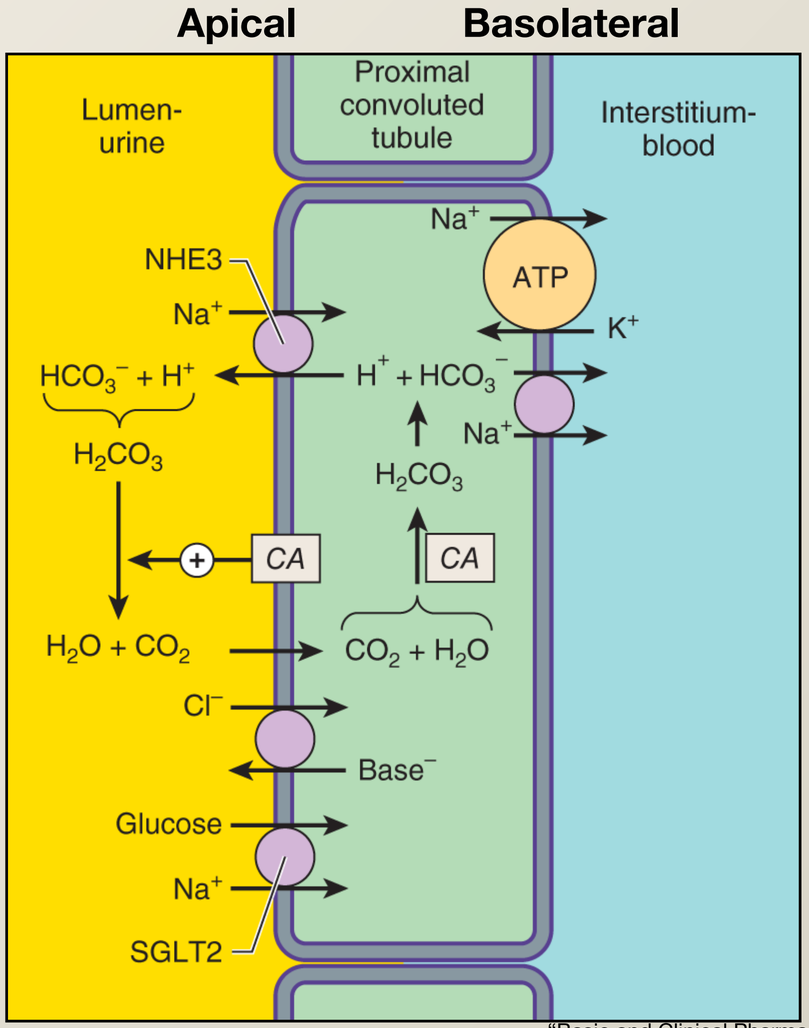
- <u>Major reabsorption activities:</u>
  - 65% of filtered **$Na^+/K^+/Ca^{2+}$ and Mg^{2+}**
  - 85% of **$NaHCO_3$**
  - ~100% of **glucose** and **amino acids**
  - Proximal tubule is **highly** permeable to water via paracellular and transcellular reabsorption

- <u>Major reabsorption mechanisms:</u>

  - Basolateral Na^+/K^+ ATPase establishes Na^+ concentration gradient

  - Secondary active transporters utilize Na^+ gradient to reabsorb glucose, amino acids, etc.

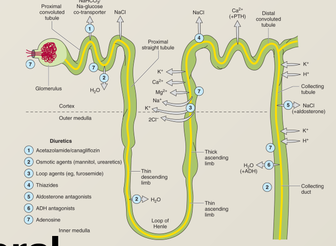
  - Carbonic anhydrase mediates **reversible** reaction to reabsorb HCO_3^-



"Basic and Clinical Pharmacology," 16e.

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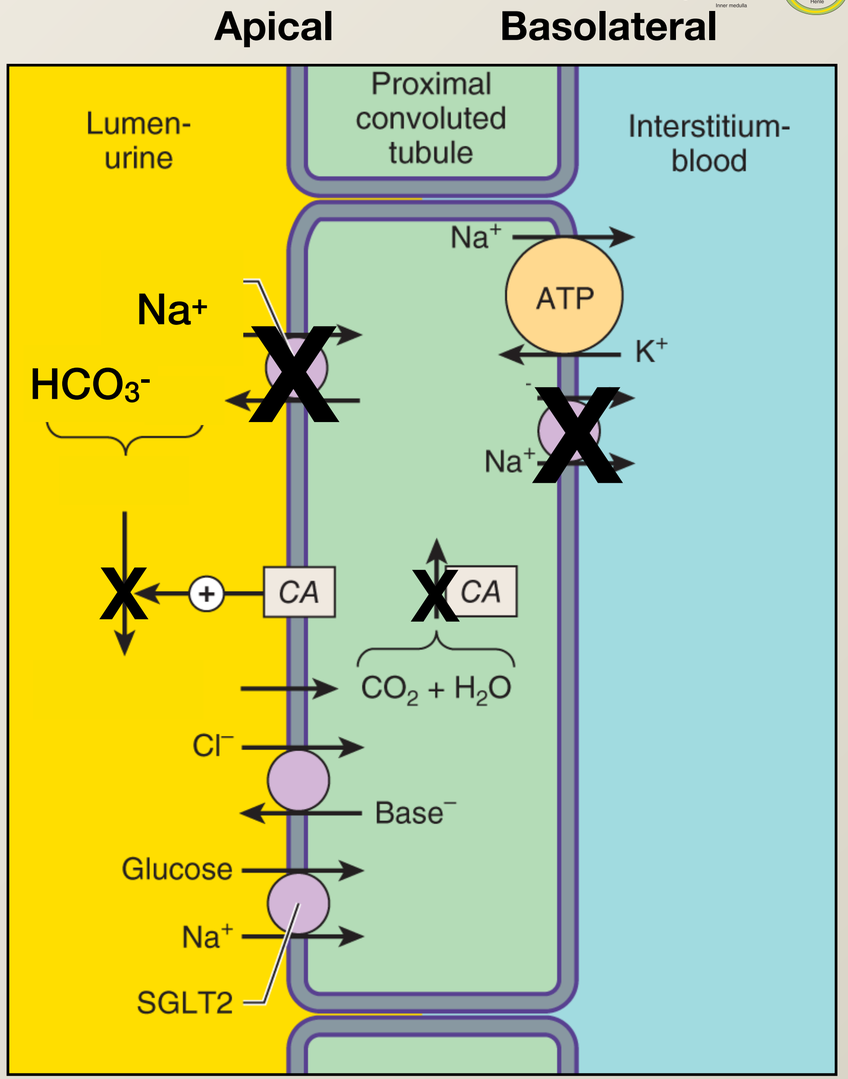
Acetazolamide: MOA



- Acetazolamide inhibits carbonic anhydrase:
  - Reduces availability of intracellular HCO_3^- and H^+

    AND

  - Reduces basolateral Na^+/ HCO_3^- symporter and apical H^+/Na^+ antiporter activity
- Net effects:
  - **Increases urinary Na^+/ HCO_3^-**
  - Water diffuses into lumen and urine output increases

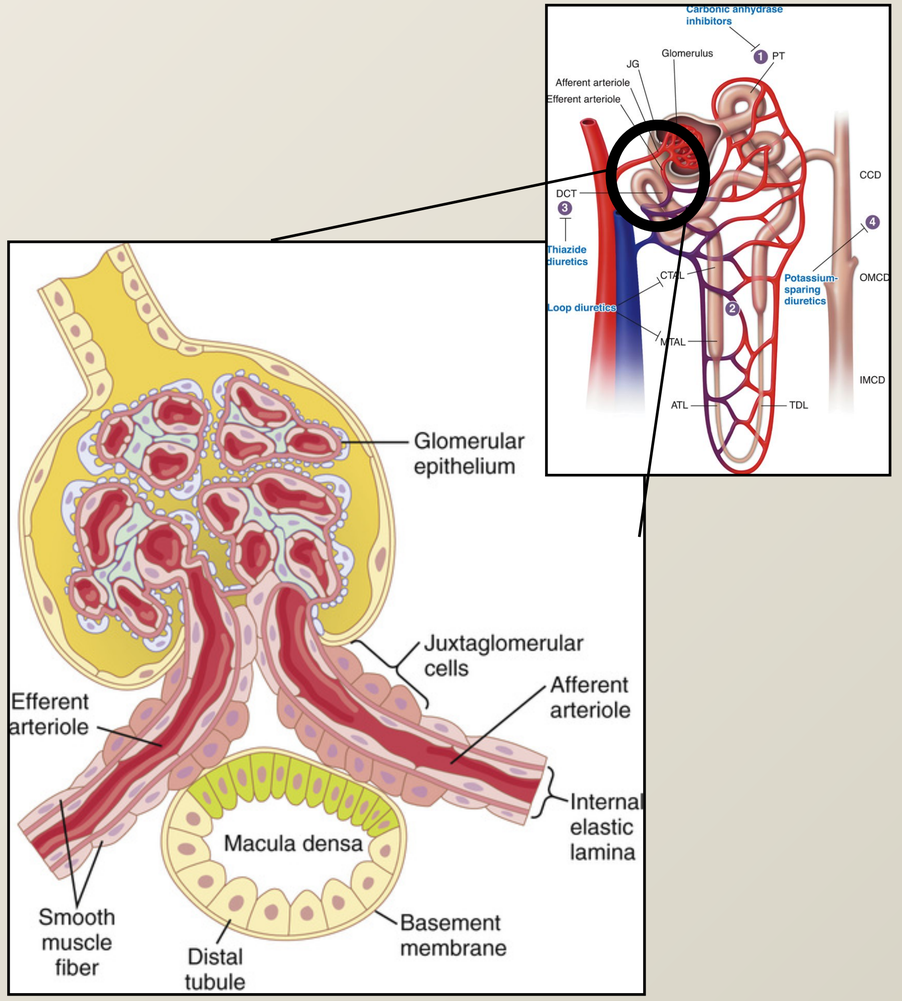


Modified from “Basic and Clinical Pharmacology,” 16e.

Review the reabsorption mechanisms in the proximal convoluted tubule (PCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics

Carbonic Anhydrase Inhibitors and Renal Hemodynamics

- Carbonic anhydrase inhibitors increase delivery of solutes to the macula densa
- Tubuloglomerular feedback increases afferent arteriolar resistance
- ***End result:*** *Carbonic anhydrase inhibitors reduce renal blood flow and GFR,* ***decreasing*** *overall strength of diuretic effect*

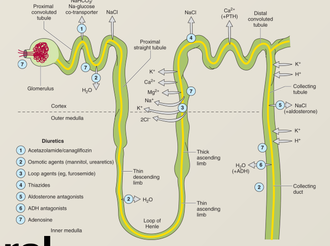


Guyton and Hall Textbook of Medical Physiology

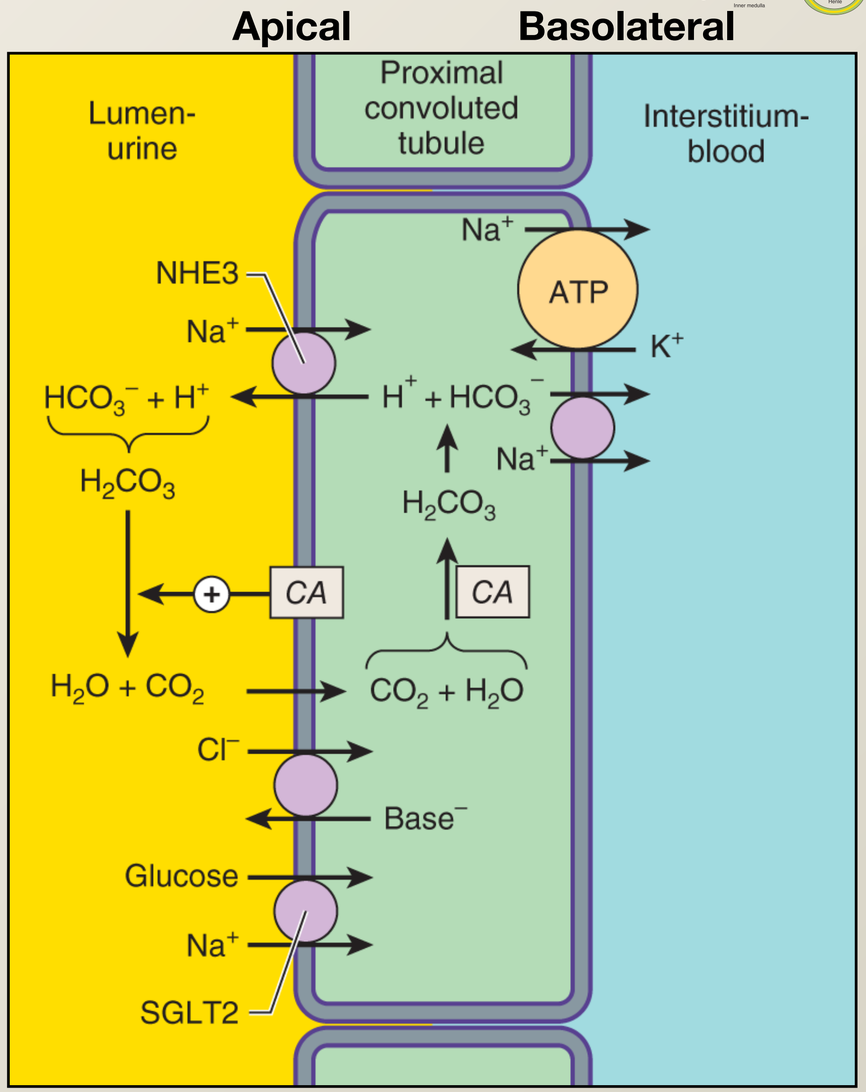
Fig. 21-5, *Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy*

Review the reabsorption mechanisms in the proximal convoluted tubule (PCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics

Acetazolamide: ADRs



- Prolonged use results in **hyperchloremic metabolic acidosis:**
  - Less HCO_3^- is reabsorbed into blood and less H^+ is secreted into urine
  - Increased Cl^- reabsorption into blood occurs in response to enhanced transepithelial electrochemical gradient
- Can cause **alkaline urine,** which may result in:
  - **phosphaturia**
  - **hypercalciuria**
  - **calcium phosphate renal stones**
- **Renal potassium wasting** due to increased Na^+ presented at collecting tubule
- **Increased risk of uricemia and gout**
  - Acetazolamide competes with uric acid secretion



"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the proximal convoluted tubule (PCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics

Acetazolamide: ADRs

- Acetazolamide may accumulate in patients with renal failure
- **Drowsiness, paresthesias** (nervous system toxicity)
  - Increased CO_2 in peripheral tissues and decreased CO_2 output in expired gas
- **Hypersensitivity reactions**
  - Fever, rashes, bone marrow suppression, interstitial nephritis
- **Contraindicated in patients with cirrhosis** due to risk of hyperammonemia and hepatic encephalopathy:
  - NH_4^+ is deprotonated in alkaline urine to NH_3, which can be reabsorbed into bloodstream

Carbonic Anhydrase Inhibitors: Other Indications

- **<u>Urinary alkalization:</u>**
  - Reduces **formation of urea/cystine crystals** in gout (may supplement with oral administration of HCO_3^-)
  - Increases excretion of organic acid anions
  - Increases excretion of salicylate (in case of overdose)

- **<u>Decreases blood/serum pH</u>**:
  - Treats metabolic alkalosis due to overcompensation of respiratory acidosis
  - Treats metabolic alkalosis due to excessive diuretic use

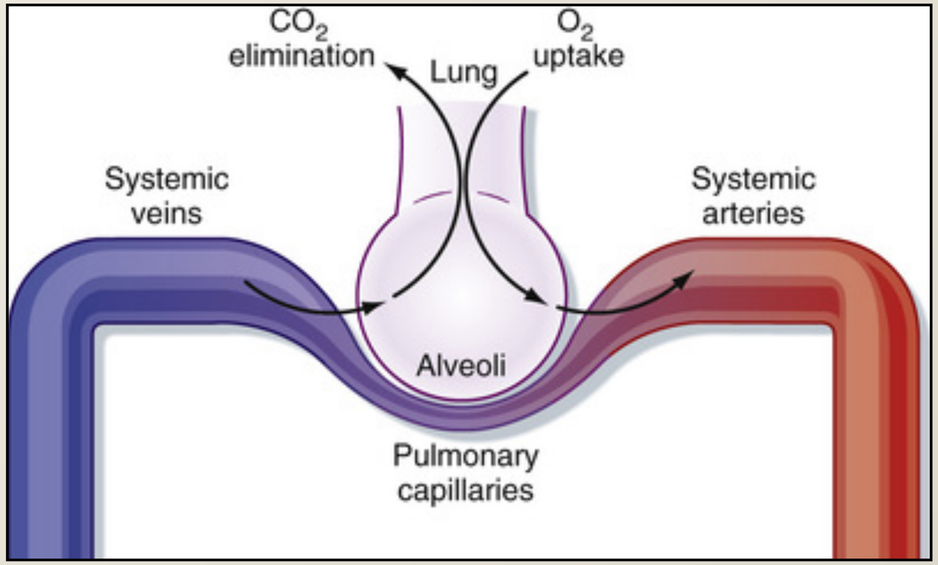
- **<u>Glaucoma</u>**
  - Inhibits carbonic anhydrase in ciliary process epithelium
  - Others indicated for *glaucoma only*:
    - Methazolamide (PO, Glauctabs), dorzolamide (topical), and brinzolamide (topical)

- **<u>Epilepsy</u>**
  - Used as adjunct agent (unknown mechanism)

- **<u>Reduction of CSF leakage</u>** (trauma, tumor, or idiopathic)

Non-diuretic Indications of Carbonic Anhydrase Inhibitors

- **Acute Mountain Sickness** (elevations over 3000m/9840ft)
- Low alveolar O_2 = low arterial O_2
  - Hyperventilation increases CO_2 elimination
    - Formation of CO_2 from H_2CO_3 removes H^+ from body and results in respiratory alkalosis
- Low alveolar O_2 can lead to hypoxic vasoconstriction
    - Pulmonary and cerebral edema
- Carbonic anhydrase inhibitors:
  - reduce loss of acid and promotes excretion of HCO_3^-
  - reduce CSF pressure and edema



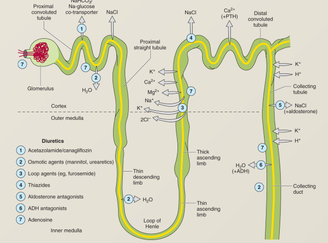
$$H^+ + HCO_3^- \rightleftharpoons H_2CO_3 \rightleftharpoons H_2O + CO_2$$

Carbonic Anhydrase

Berne & Levy, 7e

Review the reabsorption mechanisms in the proximal convoluted tubule (PCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics

Osmotic Diuretics: Mannitol and Glycerin



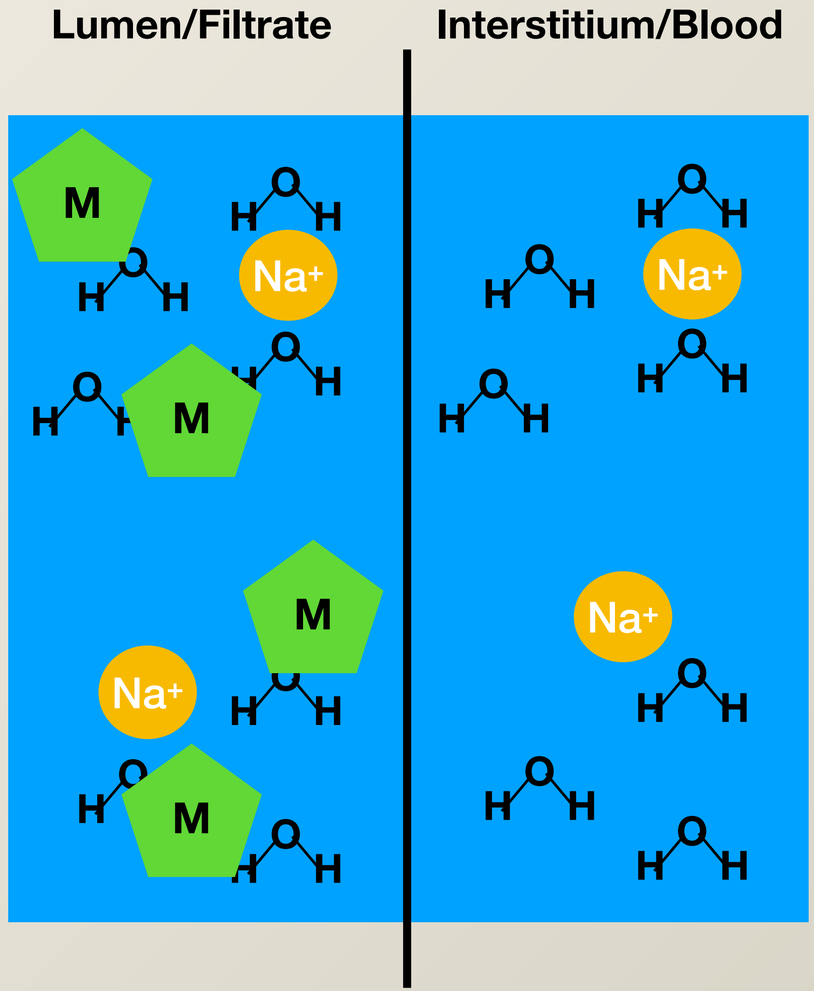
- Sites of action:
  - **Proximal tubule**
  - **Descending Limb of Loop of Henle**
  - **Collecting Duct**

- Acutely expands plasma volume, then reduces plasma volume through osmotic pressure

  - Indicated for cerebral edema, head injury, acute glaucoma

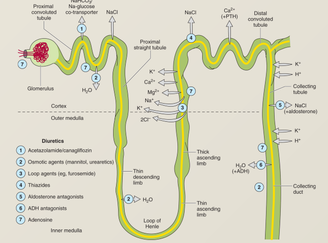
- Mannitol administered parenterally, glycerin orally

  - Filtered freely through the glomerulus, but cannot be reabsorbed



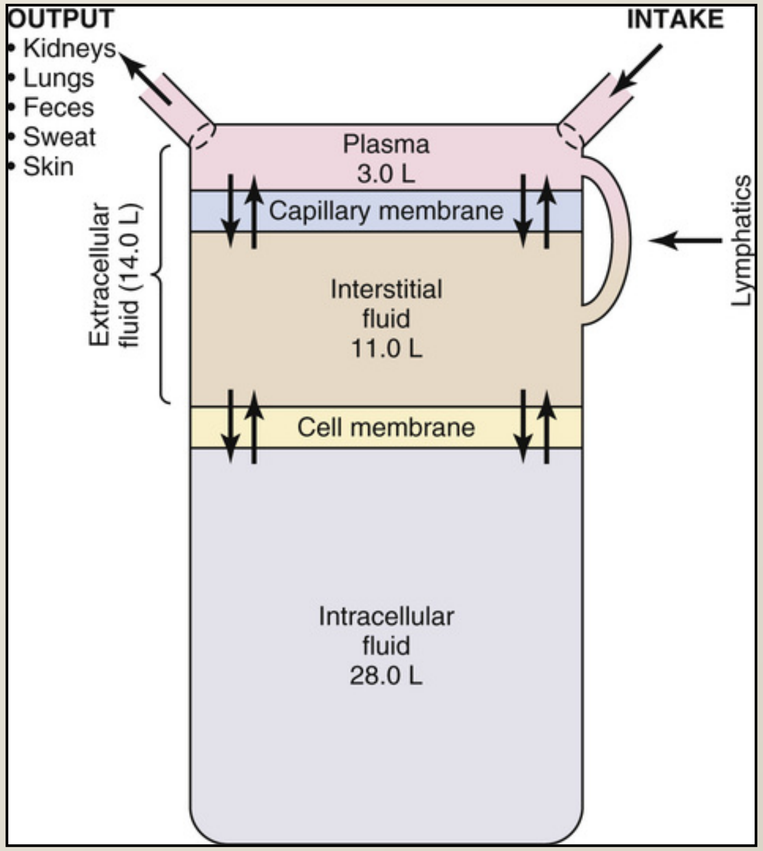
"Basic and Clinical Pharmacology," 16e.

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Osmotic Diuretics: MOA

- Two-step mechanism:
  - **First step (prediuresis):**
    - Acutely and transiently expands the body's plasma volume by "pulling" fluid from interstitial compartments
  - **Second step (postdiuresis):**
    - "Fluid pulling" increases urine output, decreasing overall ECF (plasma and interstitium)
    - Decreased plasma fluid volume causes more "pulling" of fluid from interstitial compartments (and even intracellular compartments in extreme cases)
    - Body volume is reduced through osmotic pressure
- Indicated for cerebral edema, head injury, acute glaucoma



Guyton and Hall Textbook of Medical Physiology, Figure 25-1

"Basic and Clinical Pharmacology," 16e.

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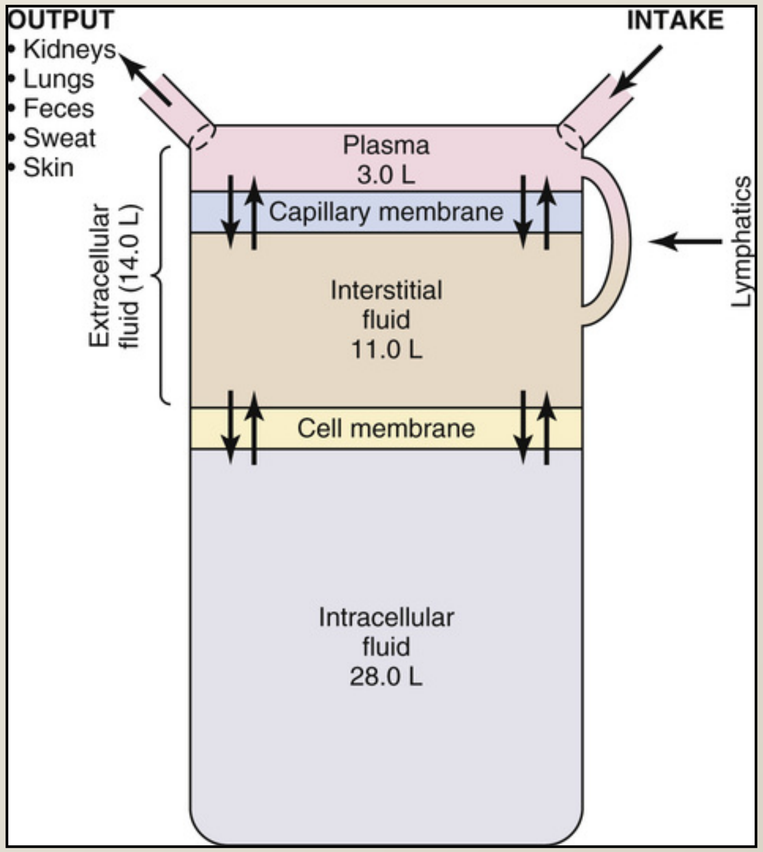
Osmotic Diuretics: ADRs

- <u>Pre-diuresis effects:</u>
  - Excess ECF (plasma) volume can cause **hyponatremia**
    - Headache, nausea, vomiting

- <u>Post-diuresis effects:</u>
  - **Hypernatremia,** dehydration, hypovolemia, hypotension, angina-like chest pain, headache
  - Acute kidney injury due to poor perfusion
  - **Hyperkalemia**
    - As ICF decreases, K^+ concentration inside cells increases

- Careful monitoring of clinical volume status and serum electrolytes is warranted

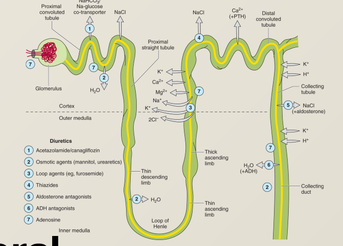
- **Contraindications**
  - Pulmonary edema, CHF (due to initial rapid volume expansion)
  - Renal failure/marginal renal function



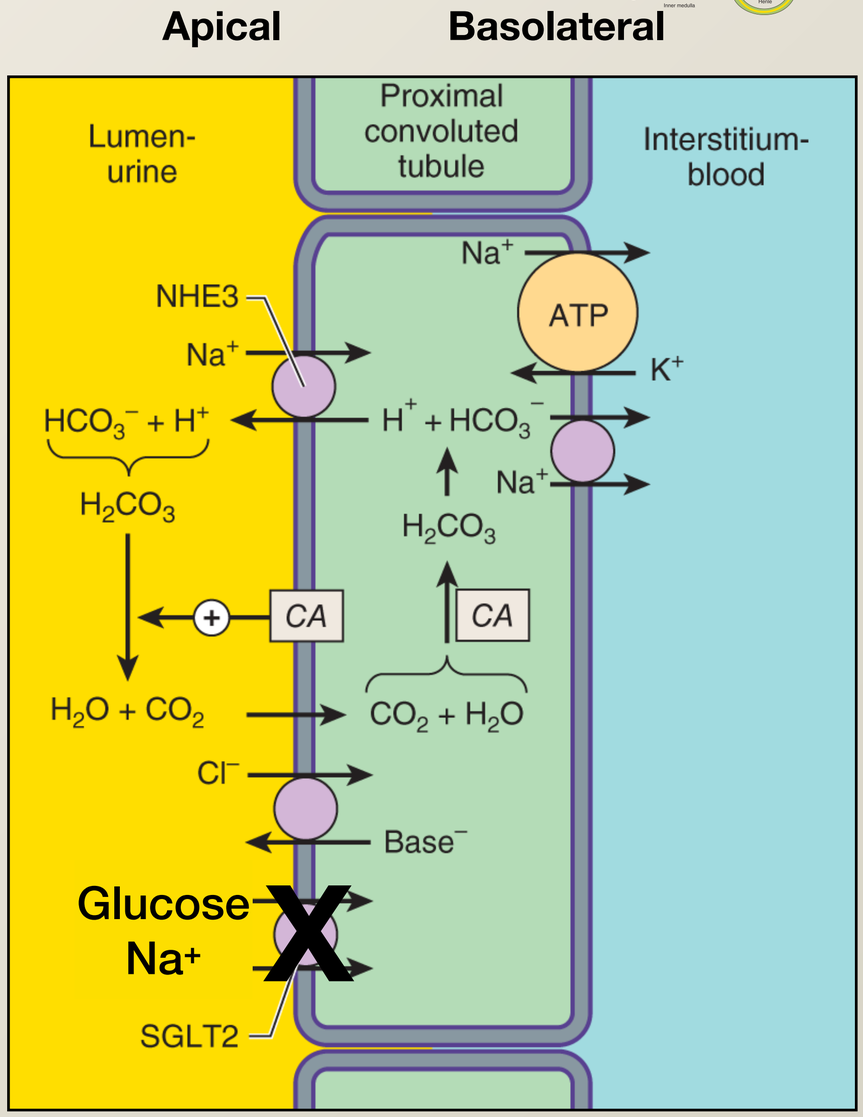
Guyton and Hall Textbook of Medical Physiology, Figure 25-1

Review the reabsorption mechanisms in the proximal convoluted tubule (PCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics

SLGT2 inhibitors



- Sodium-glucose cotransporter (SGLT2) inhibitors
  - Dapaglifozin, canaglifozin, empaglifozin
  - Indicated for diabetes mellitus
- Diuretic properties due to increased sodium and glucose excretion
- Increased glucose in urine
  - 6x increase in risk of genital fungal infection in women and slightly higher risk of UTIs
- Not recommended in renal failure



Modified from “Basic and Clinical Pharmacology,” 16e.

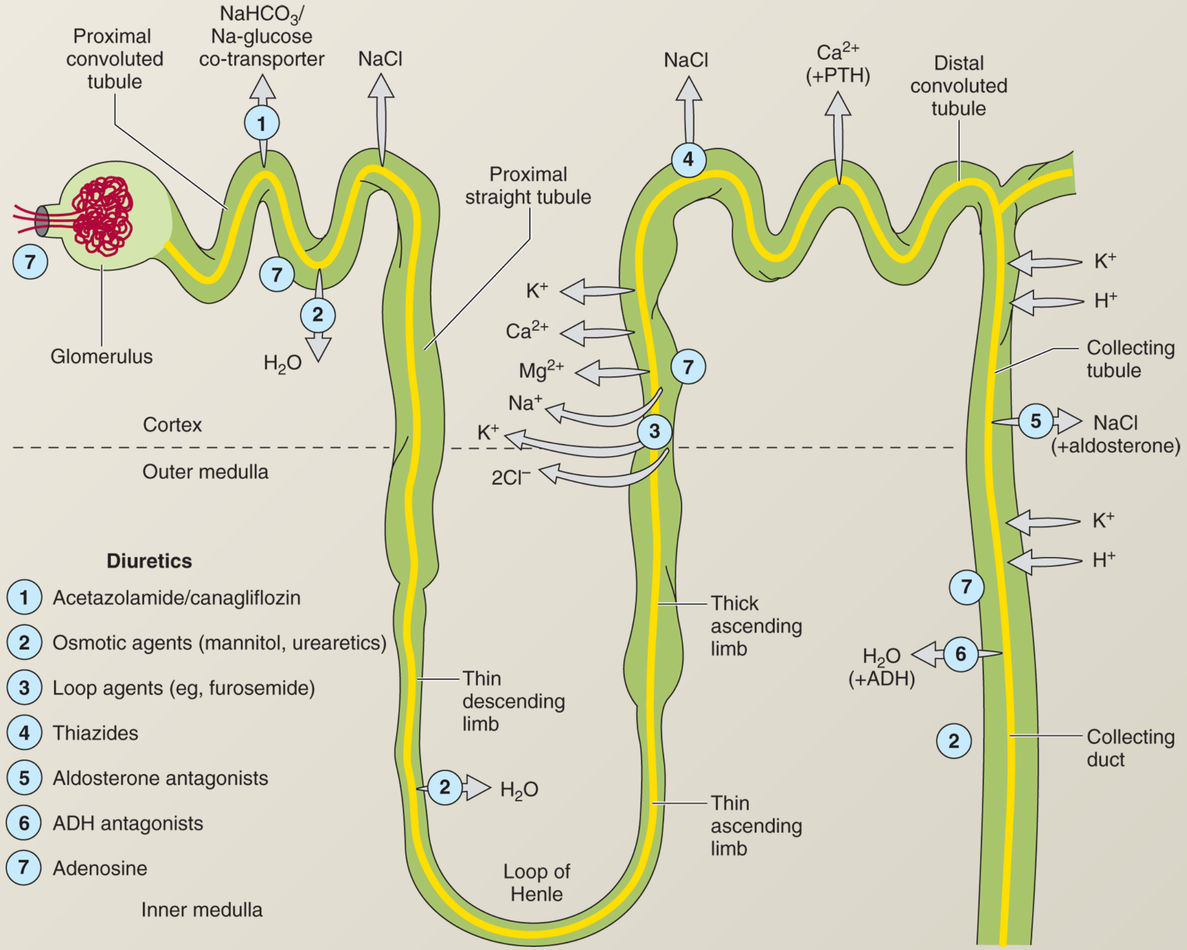
Review the reabsorption mechanisms in the proximal convoluted tubule (PCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics

Section 2: Agents Acting in the Loop of Henle, Distal Convoluted Tubule, the Cortical Collecting Tubule

- Objectives:
  - Apply physiological reabsorption mechanisms in the Loop of Henle (LOH) to discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of loop diuretics
  - Apply physiological reabsorption mechanisms in the distal convoluted tubule (DCT) to discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of thiazides
  - Apply physiological reabsorption mechanisms in the cortical collecting tubule (CCT) to discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of potassium-sparing diuretics
  - Explain the rationale behind combination therapy with loop diuretics and thiazides

Loop Diuretics

- **Furosemide** (Lasix), **bumetanide** (Bumex), **torsemide** (Demadex)
- Site of action: **Thick Ascending Limb (TAL) of the Loop of Henle**
- **<u>Powerful and fast-acting diuretics:</u>**
  - Rapid diuresis occurs within hours, may trigger antinaturesis
  - Repeated dosing, diet modification, or combination therapy required for long-term efficacy
- **Indications:**
  - Acute pulmonary/peripheral edema in HF
  - Malignant HTN (acute hypertensive crisis)
  - Volume-based HTN in CKD patients
  - Combination with thiazides for diuretic resistant essential HTN



"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the proximal convoluted tubule (PCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics

Loop Diuretics: DMPK

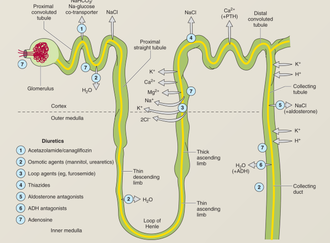
- Available IV and PO
- Rapidly absorbed, enters tubule by glomerular filtration and tubular secretion
- Eliminated by renal elimination **and hepatic metabolism**
- Half-life depends on renal function, frequent dosing is usually required

DRUG	RELATIVE POTENCY	ORAL AVAILABILITY	$t_{1/2}$ (hours)	ROUTE OF ELIMINATION
Furosemide	1	~60%	~1.5	~65% R, ~35% M[b]
Bumetanide	40	~80%	~0.8	~62% R, ~38% M
Ethacrynic acid	0.7	~100%	~1	~67% R, ~33% M
Torsemide	3	~80%	~3.5	~20% R, ~80% M
Axosemide[a]	1	~12%	~2.5	~27% R, 63% M
Piretanide[a]	3	~80%	0.6-1.5	~50% R, ~50% M
Tripamide[a]	ID	ID	ID	ID

R, renal excretion of intact drug; M, metabolism; ID, insufficient data.
[a]*Not available in the United States.*
[b]*For furosemide, metabolism occurs predominantly in the kidney.*

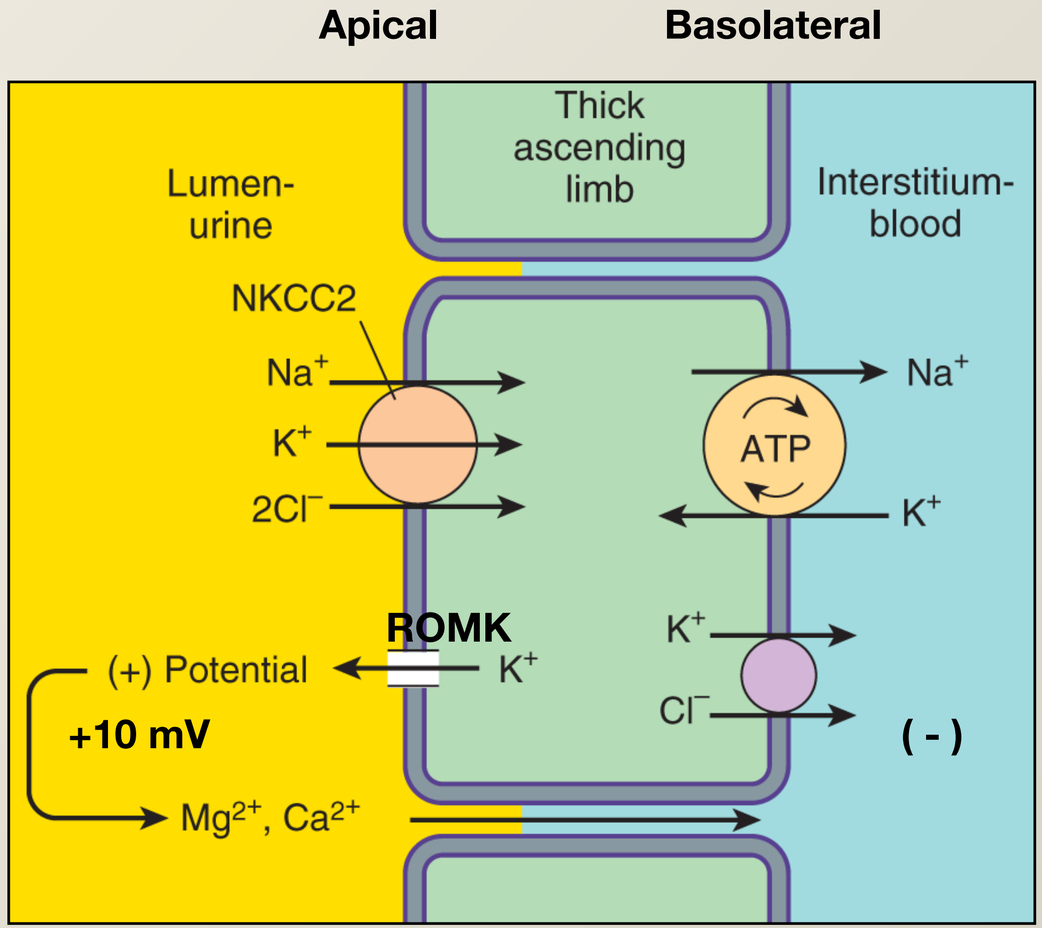
No need to memorize this table!

"Goodman & Gilman's The Pharmacological Basis of Therapeutics"



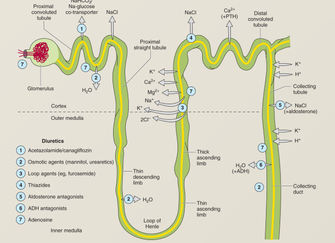
Reabsorption mechanisms in the TAL

- <u>Major reabsorption activities:</u>
  - 15–25% of filtered $Na^+/K^+/Cl^-$
  - Reabsorption of Ca^{2+} and Mg^{2+}
- TAL has **low** permeability to water, but high capacity for Na^+ reabsorption
  - *Limits efficacy of PCT diuretics*
- Basolateral Na^+/K^+ ATPase establishes Na^+ concentration gradient
- $Na^+/K^+/2Cl^-$ co transporter (NKCC2) in apical membrane transports $Na^+/K^+/2Cl^-$
- K^+ and Cl^- are transported into interstitium by K^+/Cl^- channel
- K^+ is recycled into lumen by ROMK:
  - Establishes **transepithelial voltage** differential (more positive in lumen)
  - Drives paracellular reabsorption of cations (Ca^{2+} and Mg^{2+})



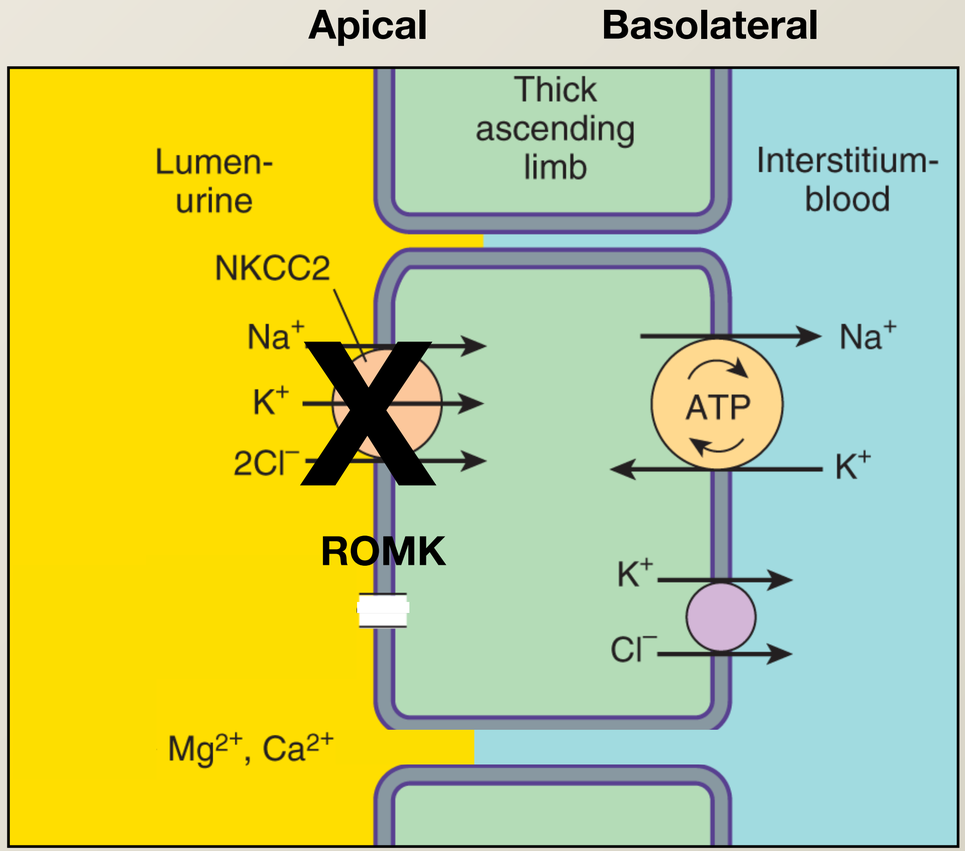
Modified from “Basic and Clinical Pharmacology,” 16e.

Review the reabsorption mechanisms in the Loop of Henle (LOH) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of loop diuretics



MOA: Loop Diuretics

- Blocks NKCC2 to reduce reabsorption of $Na^+/K^+/Cl^-$
  - Transepithelial voltage differential is **reduced**
  - Paracellular reabsorption of cations (Ca^{2+} and Mg^{2+}) is reduced
- Diuretic action of these drugs is not limited by development of acidosis
- Loop diuretics can cause renal salt wasting similar to Bartter Syndrome (TAL transporter deficits)



Modified from “Basic and Clinical Pharmacology,” 16e.

Review the reabsorption mechanisms in the Loop of Henle (LOH) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of loop diuretics

Loop Diuretics: ADRs and DDIs

- **Hypocalcemia** and **hypomagnesemia**
- **Dehydration**, hypernatremia
- **Hypokalemic Metabolic Alkalosis**
  - Loop diuretics increase Na^+ delivery to the collecting tubule, leads to enhanced secretion of K^+ and H^+
- **Hyperuricemia** (gout)
- Dose-dependent **ototoxicity**
- **Allergic reactions** (sulfonamide structure)

Drug	Common DDIs with Loop Diuretics
Aminoglycosides	Ototoxicity
Anticoagulants	Increased Anticoagulant Activity
Digitalis glycosides	Arrhythmias
Lithium	Increased Lithium levels
Propranolol	Increase plasma levels propranolol
Sulfonylureas	Hyperglycemia
Cisplatin	Ototoxcity
NSAIDs	**Blunted response and salicylate toxicity**
Probenicid	**Blunted diuretic effect**
Thiazides	**Profound diuresis**
Amphotericin B	Increased nephrotoxicity potential

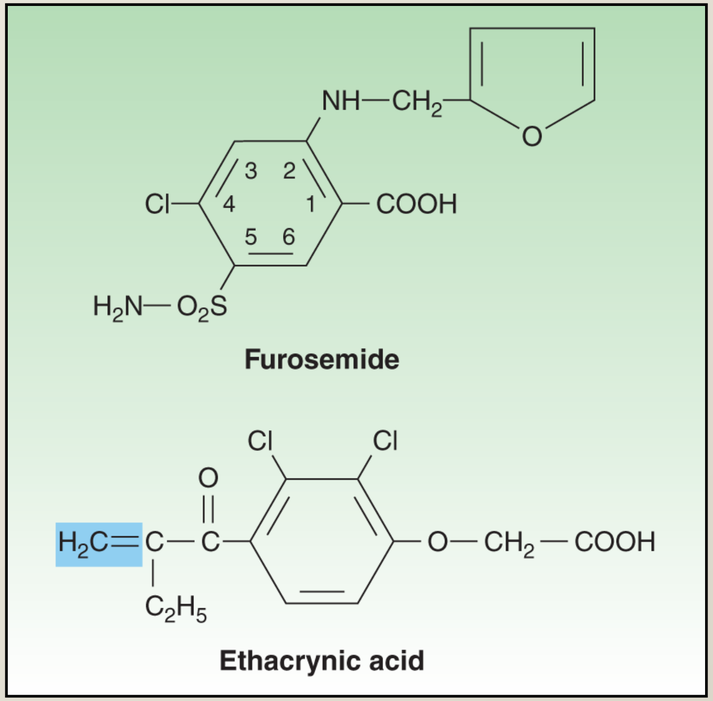
Review the reabsorption mechanisms in the Loop of Henle (LOH) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of loop diuretics

Loop Diuretics: ADRs

- *Sulfonamide*-based loop diuretics (furosemide, bumetanide, torsemide) have risk of **allergy**…

- *Phenoxyacetic* acid derivative available for patients with sulfonamide allergies:

  - **Ethacrynic acid** (Edecrin, IV)



"Basic and Clinical Pharmacology," 16th Ed.

Loop diuretics: Other effects

- Induces COX-2 expression
  - Increases prostaglandin synthesis
  - PGE2 increases renal blood flow and GFR, contributes to diuretic effect
- Effects on peripheral vascular tone make furosemide an excellent choice for **pulmonary edema and CHF edema** with venous and pulmonary congestion
- DDI with NSAIDs:
  - NSAIDs attenuate PGE2 production and secretion into tubule

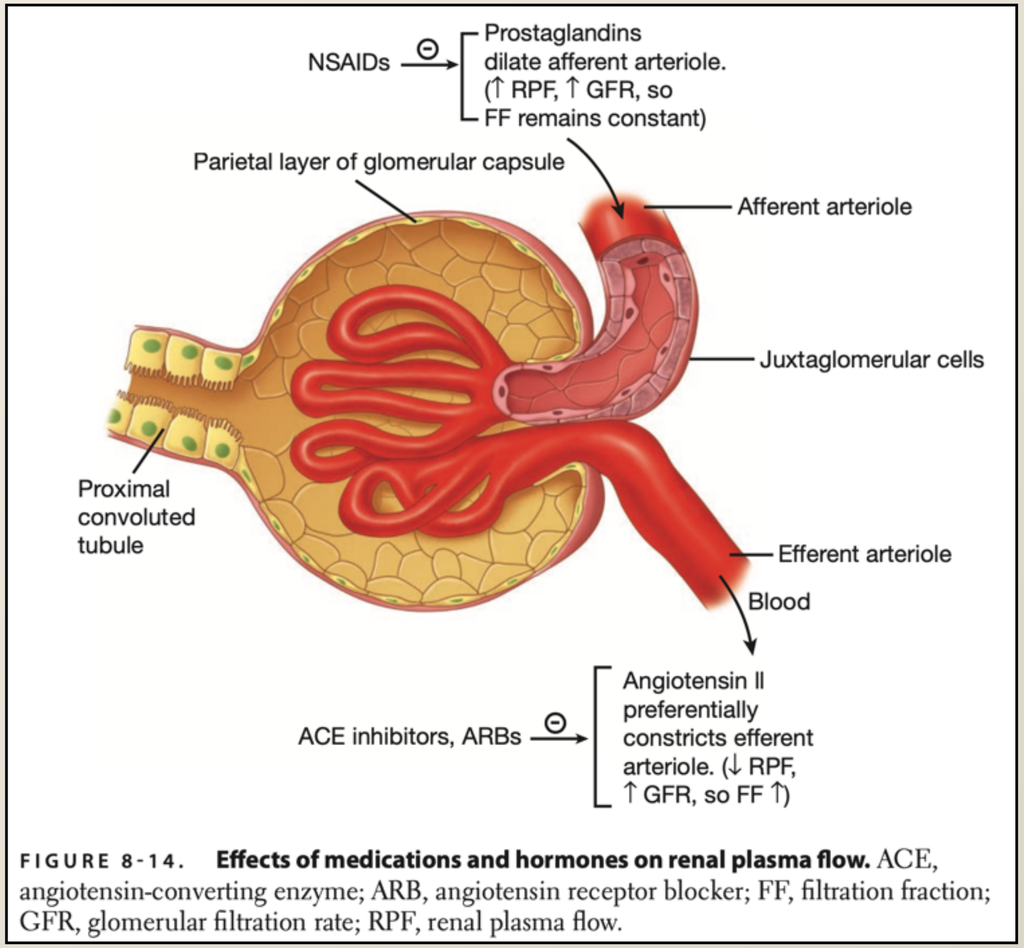


FIGURE 8-14. **Effects of medications and hormones on renal plasma flow.** ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; FF, filtration fraction; GFR, glomerular filtration rate; RPF, renal plasma flow.

Te et al., Chapter 8

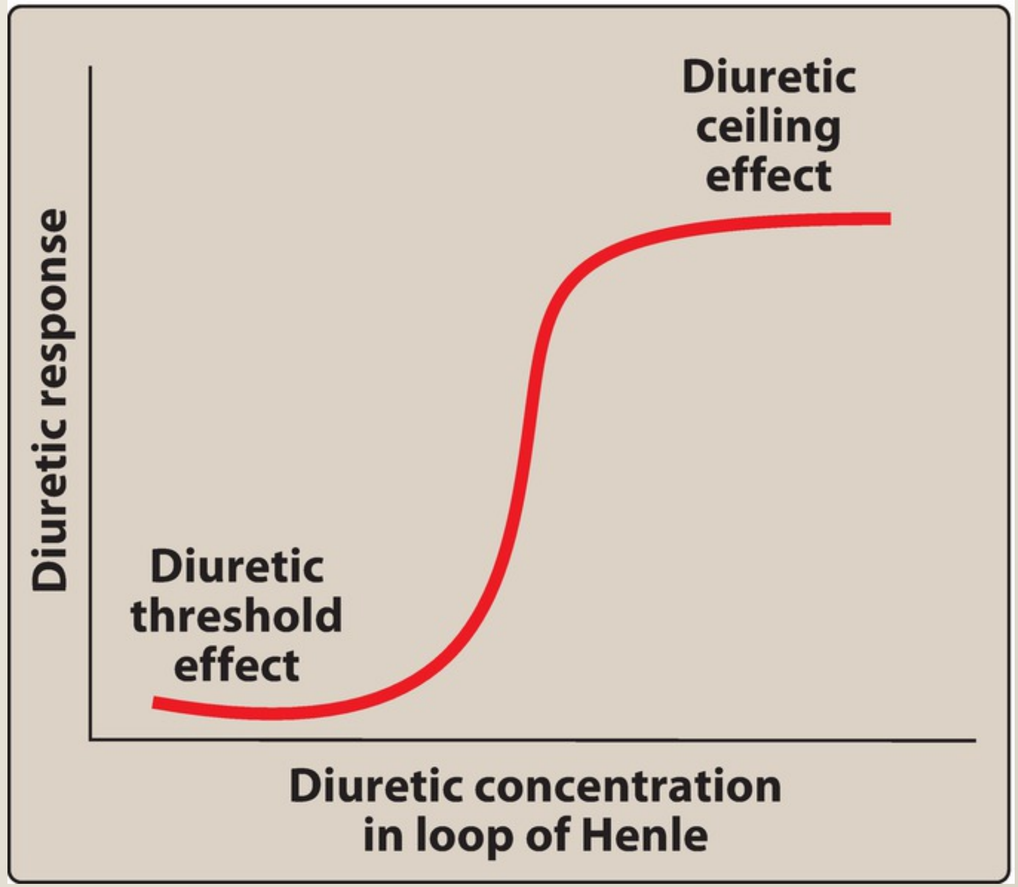
Review the reabsorption mechanisms in the Loop of Henle (LOH) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of loop diuretics

Loop Diuretics: Other Indications

- **Hypercalcemia**
- **Hyperkalemia**
  - Used in acute renal failure to enhance rate of urine flow and K^+ excretion
- **Anion overdose**
  - Bromide, fluoride, iodide
  - Must administer saline solution to replace urinary losses of Na^+ and Cl^-

Ceiling effect of Loop diuretics

- Loop diuretics must reach a threshold dose to elicit effect
- Small changes in dose causes rapid increase in diuresis
- Increasing dose will approach ceiling effect
- Effective diuretic dose varies for each patient

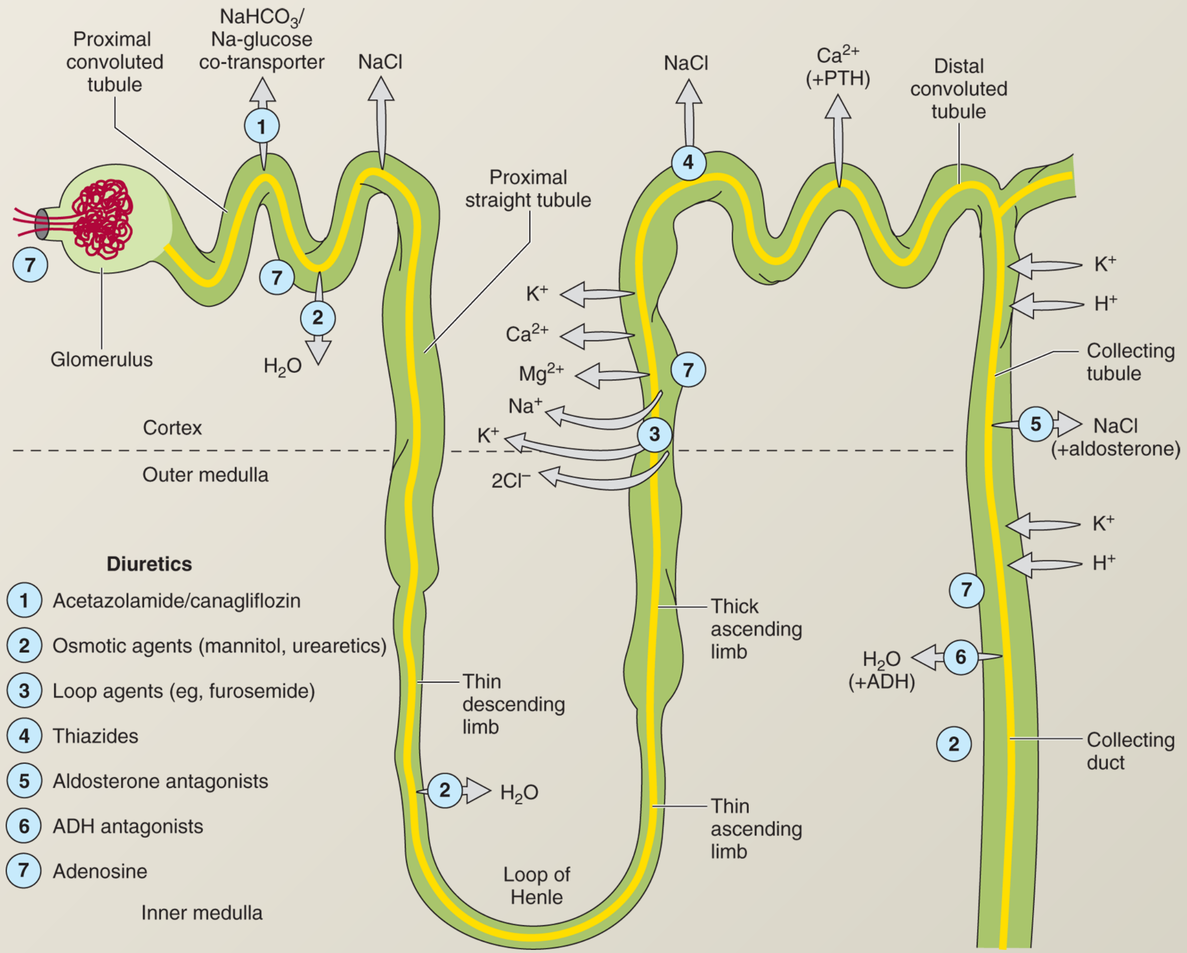


Lippincott Illustrated Reviews, 8e

Review the reabsorption mechanisms in the Loop of Henle (LOH) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of loop diuretics

Thiazides

- **Hydrochlorothiazide (HCTZ), chlorothiazide, chlorthalidone**
- Site of action: **Distal Convoluted Tubule**
- Less powerful than loop diuretics, but hypothesized to have vasodilatory effect
- **Indications:**
  - **Essential hypertension**
  - Heart failure*
  - Combination with loop diuretics for diuretic resistant essential HTN
- Thiazides are preferred for chronic and volume-based HTN
  - Elderly patients
  - Primary renal disease
  - Sodium sensitive populations



"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the proximal convoluted tubule (PCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of carbonic anhydrase inhibitors and osmotic diuretics

Thiazides DMPK

- All thiazides can be administered orally
  - Chlorothiazide is only thiazide in IV preparation
- PK and metabolic differences between members of this class
- Most are renally excreted, all are secreted into proximal tubule through organic acid secretory system

DRUG	RELATIVE POTENCY	ORAL AVAILABILITY	$t_{1/2}$ (hours)	ROUTE OF ELIMINATION
Bendroflumethiazide (NATURETIN)	10	~100%	3–3.9	~30% R, ~70% M
Chlorothiazide (DIURIL)	0.1	9–56% (dose-dependent)	~1.5	R
Hydrochlorothiazide (HYDRODIURIL)	1	~70%	~2.5	R
Hydroflumethiazide (SALURON)	1	~50%	~17	40-80% R, 20-60% M
Methyclothiazide (ENDURON)	10	ID	ID	M
Polythiazide (RENESE)	25	~100%	~25	~25% R, ~75% U
Trichlormethiazide (NAQUA)	25	ID	2.3–7.3	R
Chlorthalidone (HYGROTON)	1	~65%	~47	~65% R, ~10% B, ~25% U
Indapamide (LOZOL)	20	~93%	~14	M
Metolazone (MYKROX, ZAROXOLYN)	10	~65%	ID	~80% R, ~10% B, ~10% M
Quinethazone (HYDROMOX)	1	ID	ID	ID

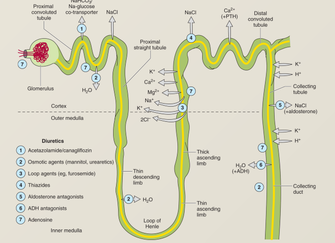
R, renal excretion of intact drug; M, metabolism; B, excretion of intact drug into bile; U, unknown pathway of elimination; ID, insufficient data.

No need to memorize this table!

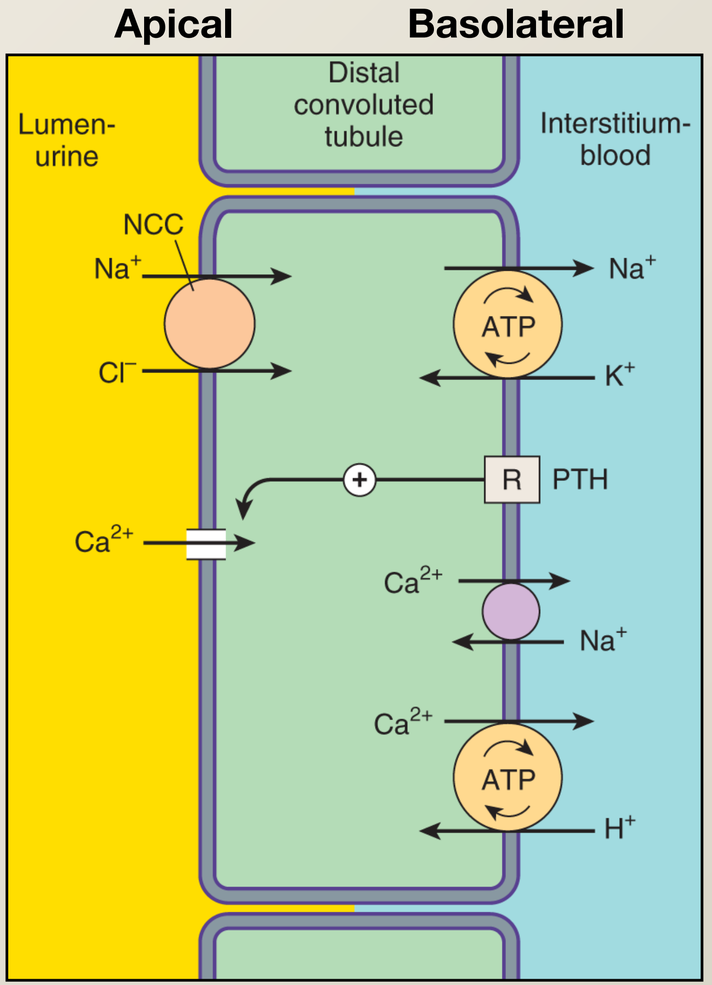
"Goodman & Gilman's The Pharmacological Basis of Therapeutics"

Review the reabsorption mechanisms in the distal convoluted tubule (DCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of thiazides

Reabsorption mechanisms in the DCT

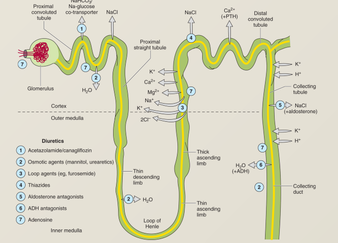


- <u>Major reabsorption activities:</u>
  - Reabsorption of 4–8% of filtered Na^+ and Cl^-
  - Ca^{2+} reabsorption under PTH control
- DCT has **low** permeability to water
  - *Tubular site of macula densa*
- Basolateral Na^+/K^+ ATPase establishes concentration gradient
- The Na^+/Cl^- co transporter (NCC) in the apical (luminal) membrane transports Na^+/Cl^-
- Ca^{2+} is transported across the basolateral membrane into the interstitium by the Na^+/Ca^{2+} exchanger and by the Ca^{2+}ATPase
- Mg^{2+} is absorbed by magnesium-selective channels and unknown mechanism



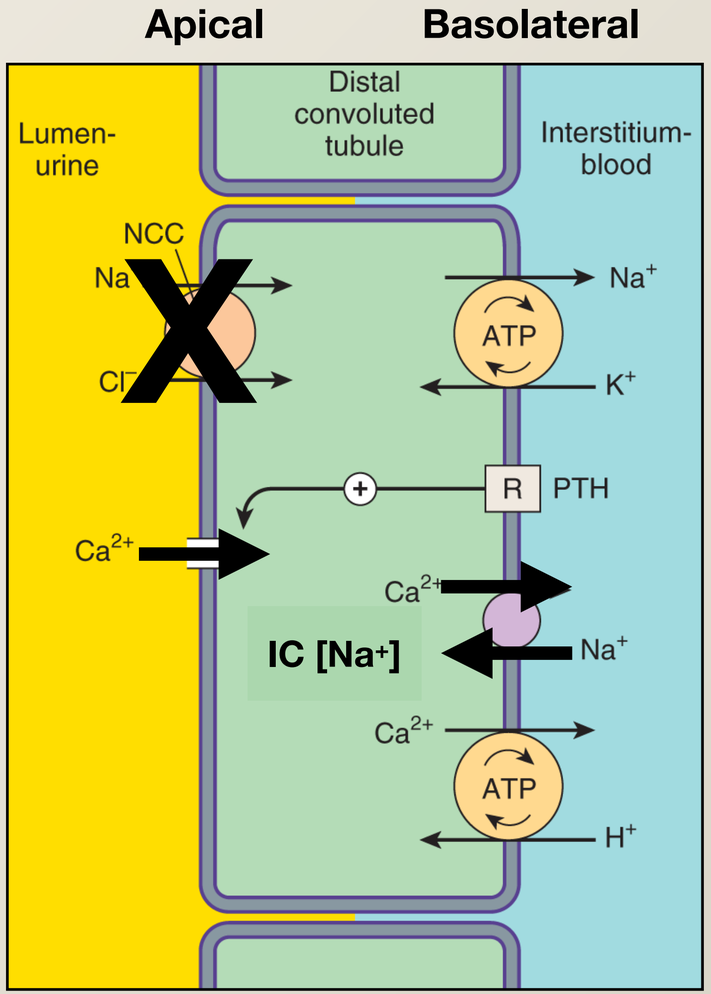
"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the distal convoluted tubule (DCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of thiazides



MOA: Thiazides

- Thiazides block the Na^+/Cl^- co-transporter (NCC)
  - *Results in lower than normal intracellular (IC) Na^+ concentration*
  - Drives basolateral Na^+/Ca^{2+} exchanger
    - More Ca^{2+} is absorbed into interstitium-blood
- Rarely causes hypercalcemia, but can unmask hypercalcemia due to other causes
  - Unknown mechanism: thiazides also increase Mg^{2+} absorption

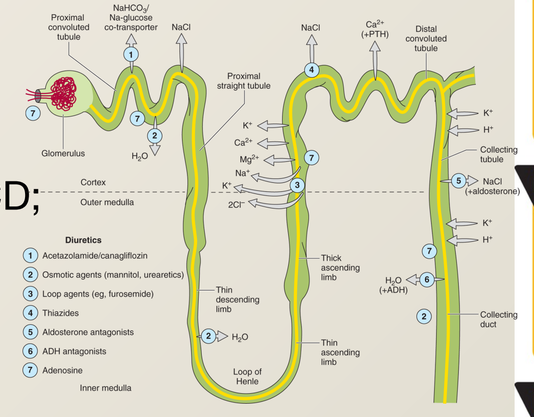


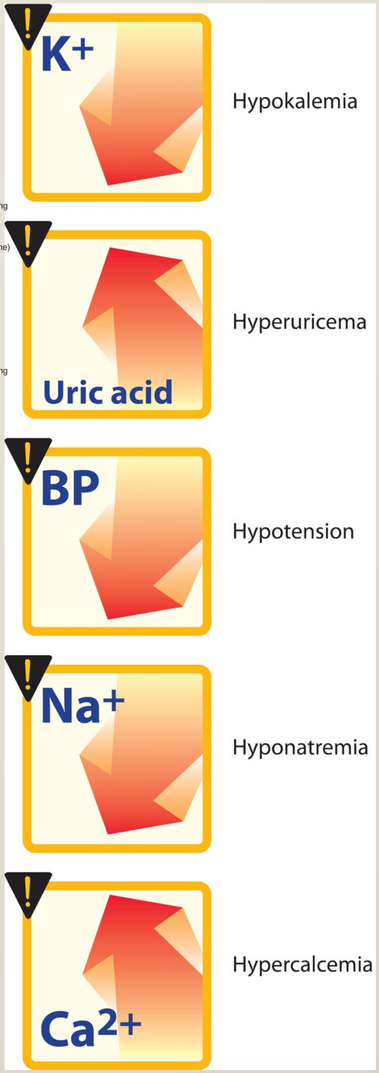
Modified from “Basic and Clinical Pharmacology,” 16e.

Review the transport mechanisms along the nephron tubule and illustrate the changes that occur with electrolyte transport and water reabsorption in response to diuretics

Thiazide ADRs

- **Hypokalemic Metabolic Alkalosis**
- **Hyponatremia**
  - Location of DCT promotes ADH-independent water reabsorption in CCD; explains contraindication in heart failure
- **Hyperuricemia**
- Weakness, fatigue, paresthesias
- Impaired Carbohydrate Tolerance (**hyperglycemia**)
  - Off-target effect to activates ATP-sensitive K^+ channels on ß-cells in pancreas
  - Inhibition of insulin release
- **Hyperlipidemia** - 5-15% increase in total serum cholesterol and LDLs
- **Contraindications:**
  - Cirrhotic patients, borderline renal failure
- **Allergic reactions** (sulfonamide)



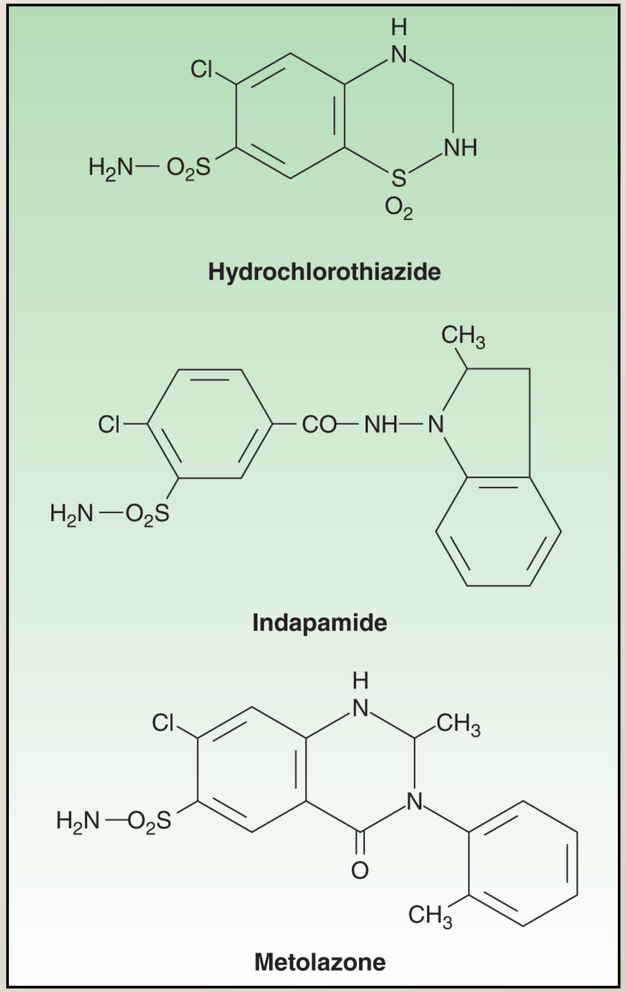


Lippincott Illustrated Reviews, 8e

Review the reabsorption mechanisms in the distal convoluted tubule (DCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of thiazides

Thiazides: Other Indications

- **Nephrogenic diabetes insipidus…**
  - *Counterintuitive mechanism…*
- **Nephrolithiasis** due to idiopathic hypercalciuria
  - Thiazides increase absorption of calcium
  - May modestly reduce the risk of osteoporotic fractures

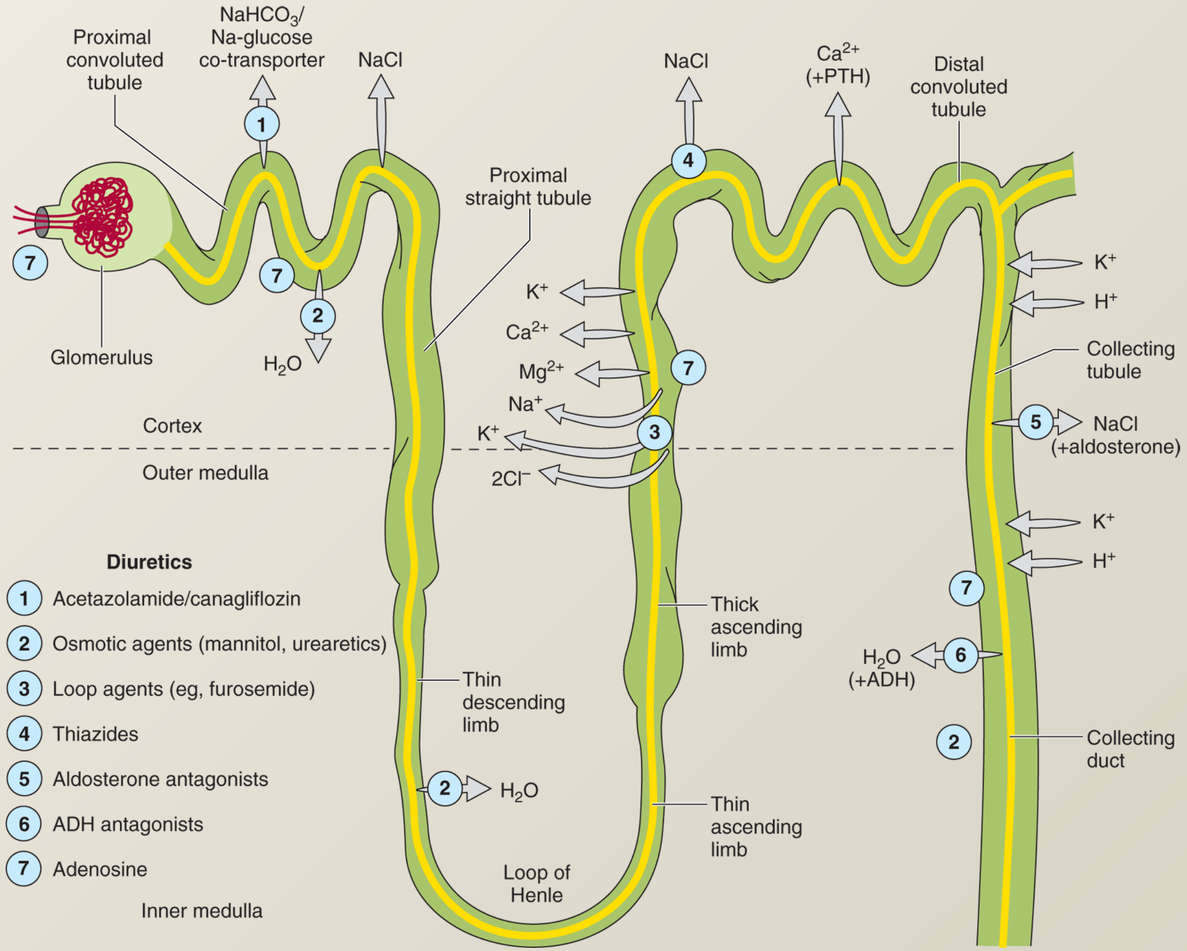


"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the distal convoluted tubule (DCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of thiazides

Potassium-Sparing Diuretics

- Class #1: Mineralocorticoid receptor antagonists:
  - **Spironolactone** (Aldactone), eplenerone (Inspra)
- Class #2: Epithelial Na^+ channel (ENaC) blockers:
  - **Amiloride** (Midamor)
  - Triamterene (Dyrenium)
- Site of action: **Cortical Collecting Tubule**
- Antagonize effects of aldosterone
- <u>Indications:</u>
  - **Addition to other diuretics to reduce K^+ wasting**
  - Rarely used as monotherapy

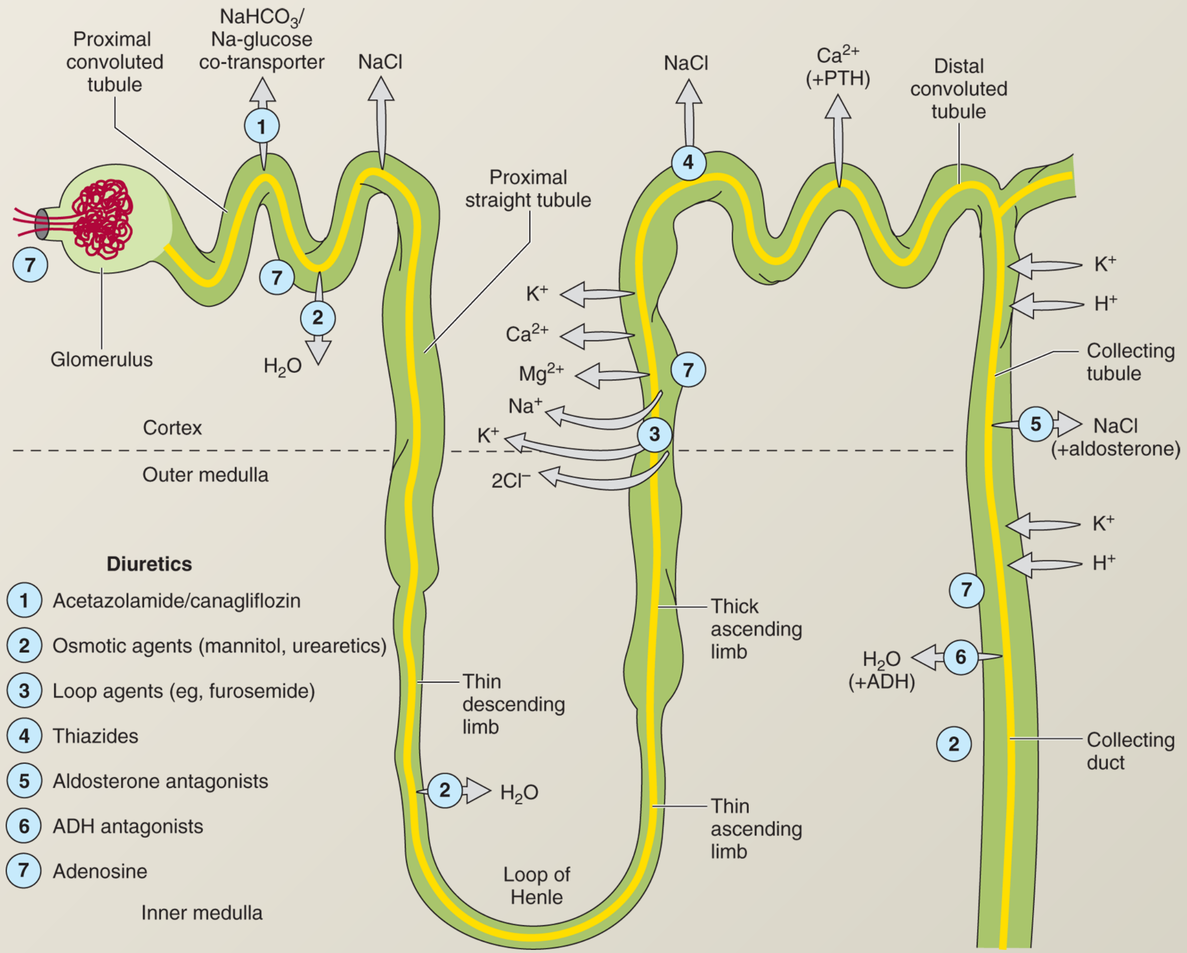


"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the cortical collecting tubule (CCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of potassium-sparing diuretics

Potassium-Sparing Diuretics

- Mineralocorticoid receptor antagonists are preferred for conditions with high aldosterone
    - HTN due to hyperaldosteronism, cirrhosis, heart failure
    - Indicated for resistant hypertension
    - Protects against restructuring of the heart
- ENaC blockers preferred for polyuria, polydipsia (lithium-induced nephrogenic diabetes insipidus)



"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the cortical collecting tubule (CCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of potassium-sparing diuretics

DMPK

- Administered PO
- Secreted into the proximal tubule
- Most K^+ sparing drugs are hepatically metabolized

DRUG	ORAL AVAILABILITY	$t_{1/2}$ (hours)	ROUTE OF ELIMINATION
Spironolactone	~65%	~1.6	M
Canrenone[a]	ID	~16.5	M
Potassium canrenoate[a]	ID	ID	M
Eplerenone	ID	~5	M

M, metabolism; ID, insufficient data.
[a]Not available in United States.

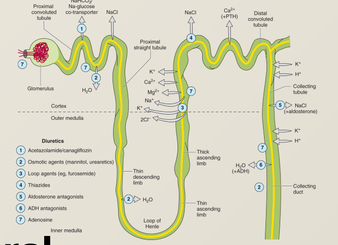
DRUG	RELATIVE POTENCY	ORAL BIOAVAILABILITY	$t_{1/2}$ (hours)	ROUTE OF ELIMINATION
Amiloride	1	15–25%	~21	R
Triamterene	0.1	~50%	~4	M

R, renal excretion of intact drug; M, metabolism; however, triamterene is transformed into an active metabolite that is excreted in the urine.

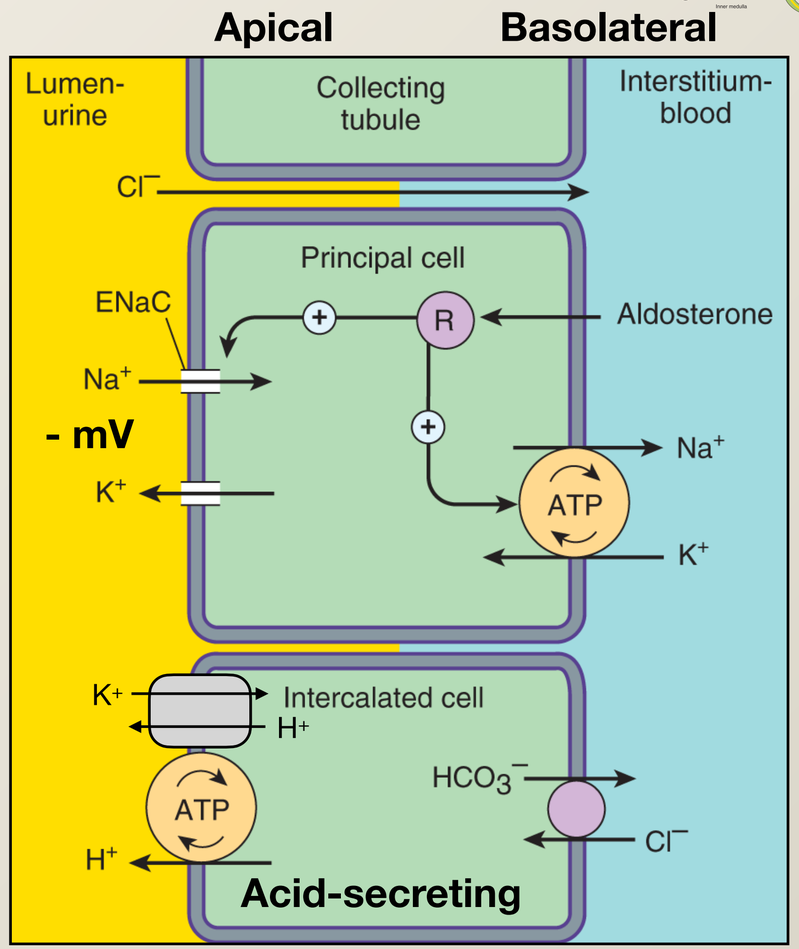
No need to memorize this table!

"Goodman & Gilman's The Pharmacological Basis of Therapeutics"

Reabsorption mechanisms in the CCT



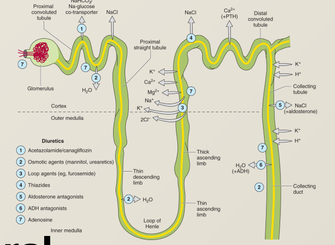
- <u>Major reabsorption activities:</u>
  - Na^+ reabsorption (2–5%)
  - Fine-tuning of K^+ and H^+ secretion
- CCT has **variable** water reabsorption, continuous with CCD
- Na^+/K^+ ATPase establishes concentration gradient
- ENaC transports Na^+ into cell, creates negative transepithelial potential
  - K^+ is excreted into lumen, drives H^+ secretion by intercalated cells
  - Cl^- is absorbed, drives H^+ secretion by intercalated cells
- Low volume triggers RAAS:
  - Aldosterone binds to mineralocorticoid receptor, promotes delivery of ENaC to apical membrane
- Other diuretics cause hypokalemia due to increased Na^+ presented at collecting tubule



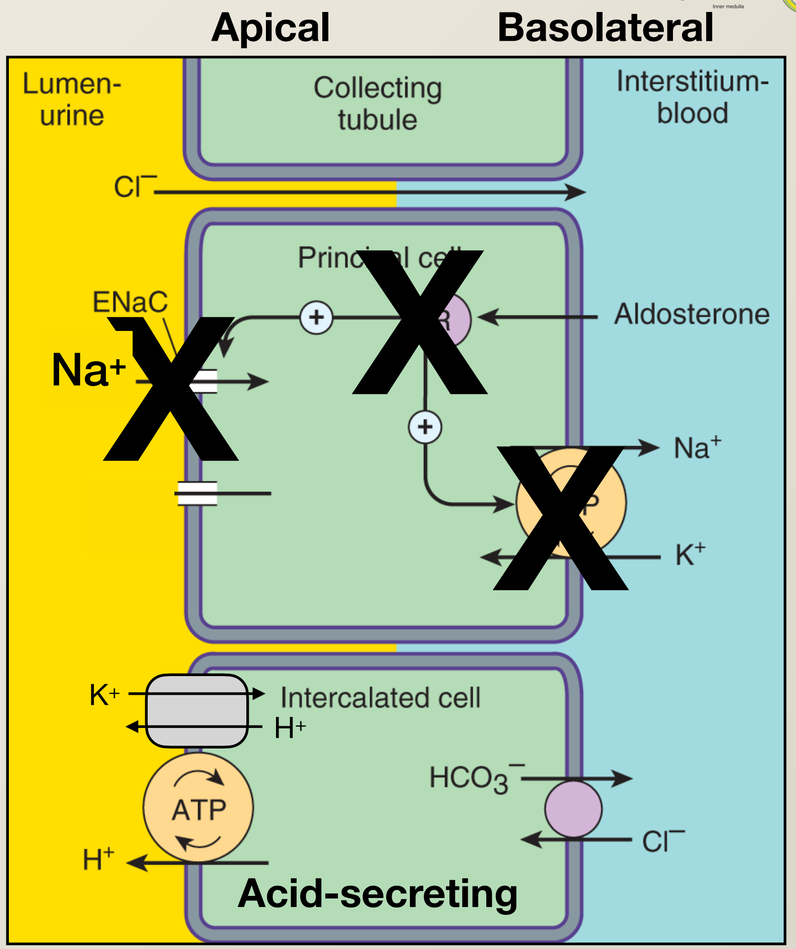
"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the cortical collecting tubule (CCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of potassium-sparing diuretics

MOA: K+ Sparing Diuretics



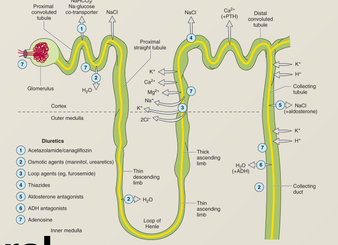
- **Spironolactone** inhibits aldosterone receptor
  - Reduces ENaC present at apical membrane



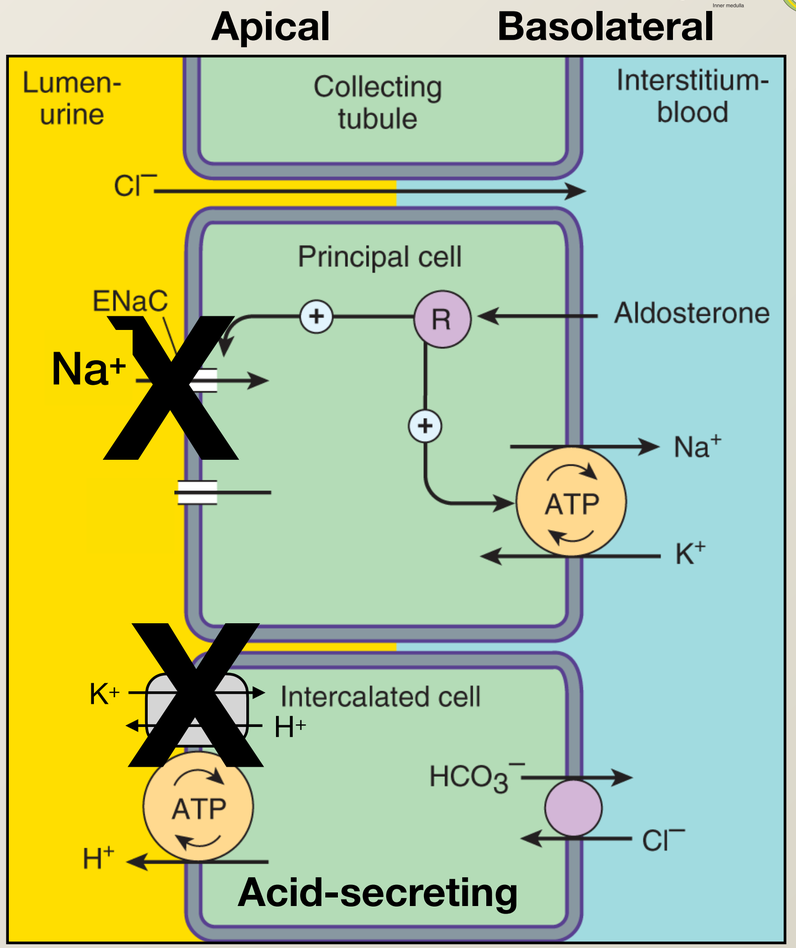
Modified from “Basic and Clinical Pharmacology,” 16e.

Review the reabsorption mechanisms in the cortical collecting tubule (CCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of potassium-sparing diuretics

MOA: K^+ Sparing Diuretics

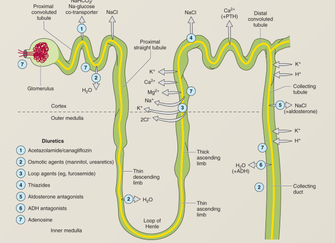


- **Amiloride** blocks ENaC, prevents negative transepithelial potential
- Neither promote increased potassium excretion
- Can reduce H^+ secretion



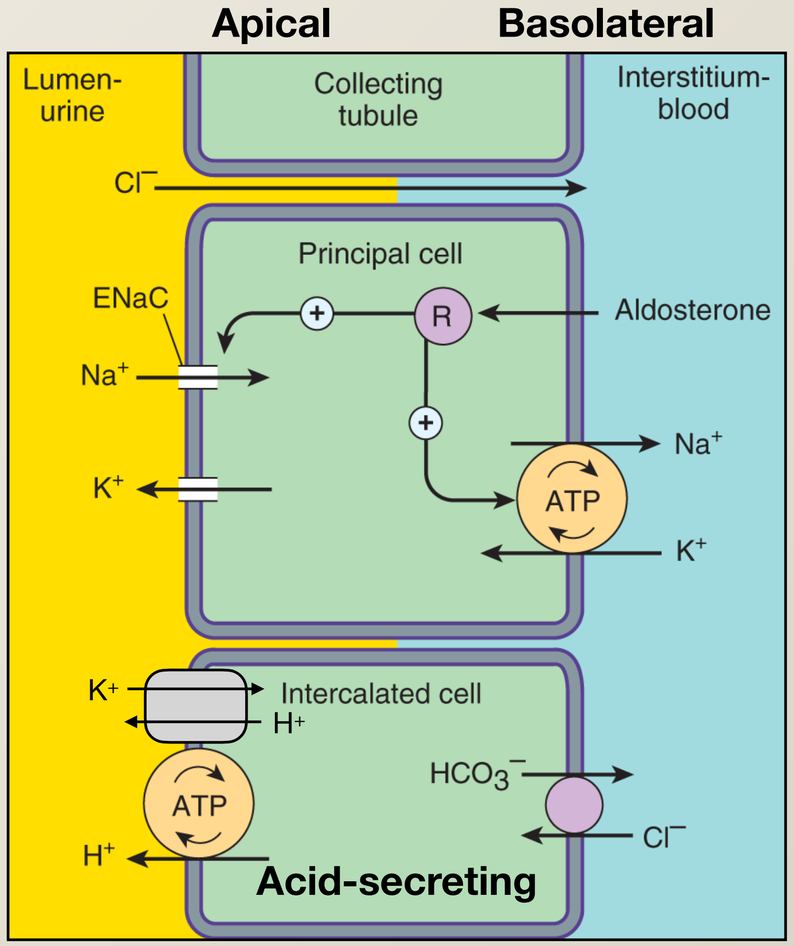
Modified from “Basic and Clinical Pharmacology,” 16e.

Review the reabsorption mechanisms in the cortical collecting tubule (CCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of potassium-sparing diuretics



Toxicities of K^+ Sparing Diuretics

- **Hyperkalemia**
  - Risk increased by use of other drugs that reduce or inhibit renin (ß blockers, NSAIDs) or ACE inhibitors
  - Combinations of K-sparing and thiazide diuretics reduce thiazide-induced hypokalemia
    - Metabolic alkalosis is ameliorated
  - Contraindicated in patients with chronic renal insufficiency
- **Hyperchloremic Metabolic Acidosis** due to inhibition of H^+ secretion
- Spironolactone: **gynecomastomia**



"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the cortical collecting tubule (CCT) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of potassium-sparing diuretics

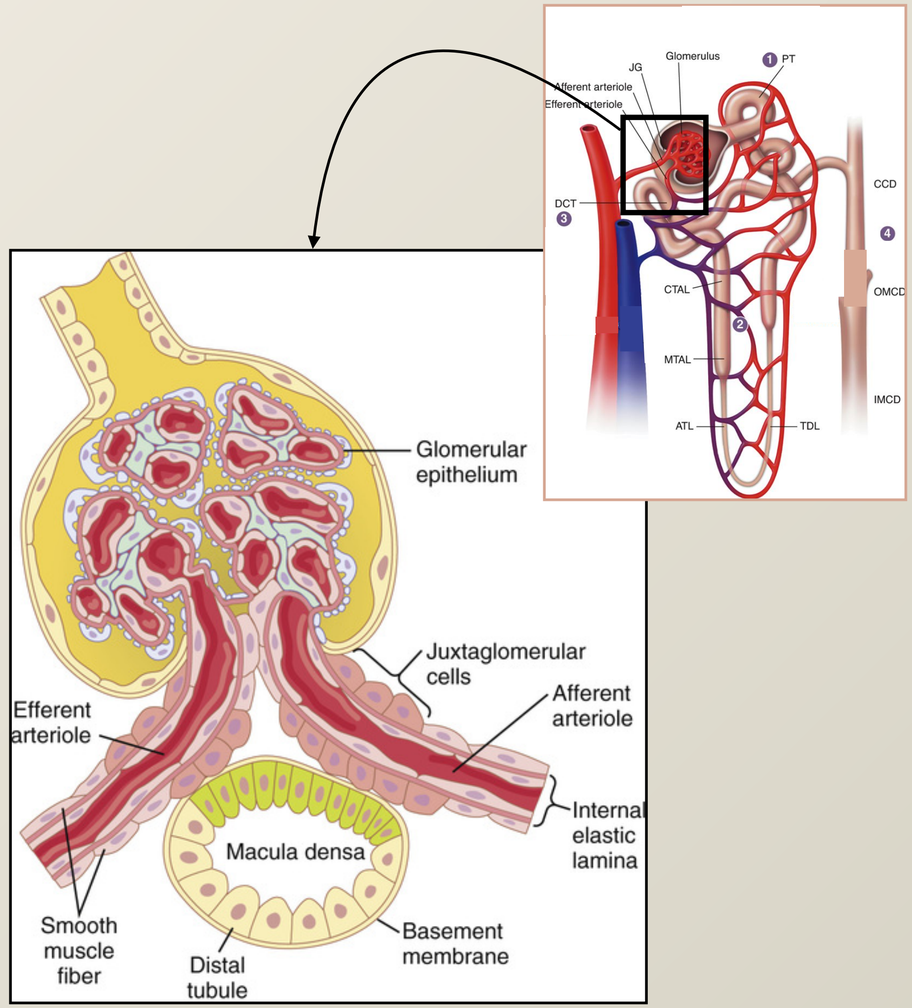
Indications of K^+ Sparing Diuretics

- Potassium-sparing diuretics are most useful in **hyperaldosteronism**
  - Primary hypersecretion (Conn's syndrome, ectopic adrenocorticotropic hormone production)
  - Secondary hyperaldosteronism due to diminished intravascular volume (heart failure, hepatic cirrhosis, nephrotic syndrome)
- Reduced **K^+ secretion due by other diuretics**
- Liddle's syndrome (amiloride)
  - Rare autosomal dominant disorder that results in activation of sodium channels in the cortical collecting ducts

Review the reabsorption mechanisms in the cortical collecting duct (CCD) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of potassium-sparing diuretics

Common compensatory mechanisms in response to diuretic therapy

- Strong natriuresis by loop diuretics may trigger **RAAS** and promote sodium reabsorption (antinatriuresis)
  - **“Postdiuretic NaCl retention”**
    - Patients on diuretic therapy should be placed on low-sodium diet
    - Dosage frequency and amount should be considered
- **Renal hemodynamics** and **tubuloglomerular feedback (TGF)**
  - *“Changes in ionic composition in one tubule segment alters ionic composition presented to distal regions”*
  - GFR can be altered by feedback from the macula densa
- Chronic treatment with diuretics may result in tubular remodeling (hypertrophy)
  - Combination therapy may be considered for treatment-resistant conditions



Guyton and Hall Textbook of Medical Physiology

Fig. 26-5, *Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy,* 5e

Diuretic adaptation and resistance

- Kidney is highly adaptive
- Initial dose: brisk diuresis
- “Braking phenomenon” occurs in 1-2 days
- Diuretic tolerance can occur due to compensatory mechanisms or **physiologic adaptations**

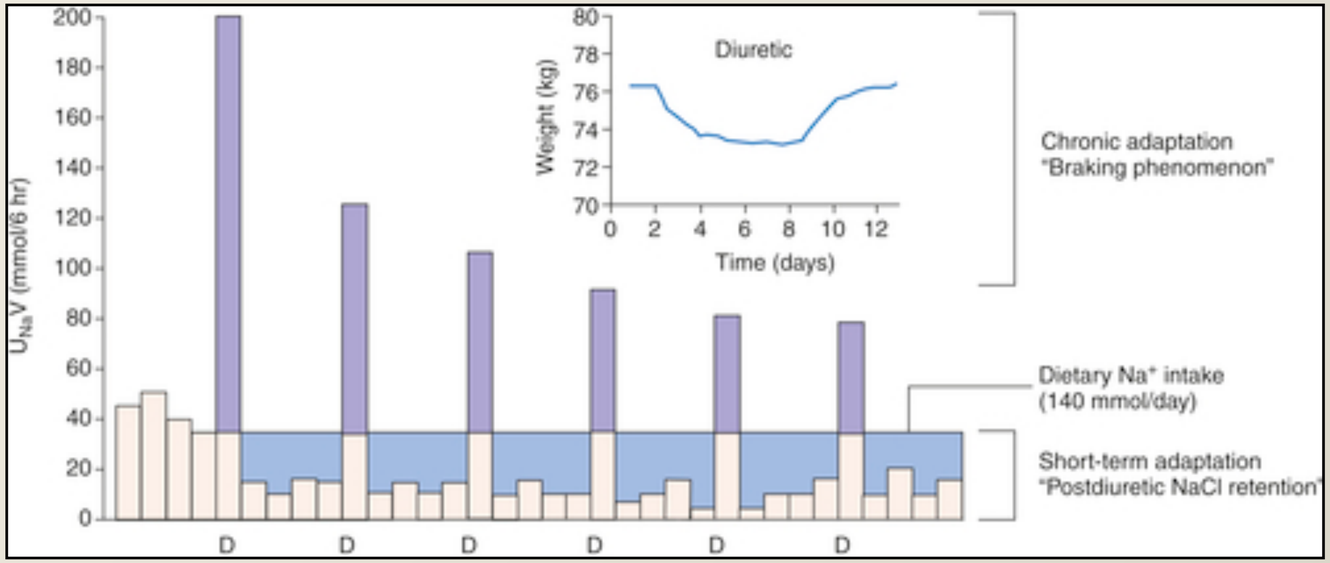


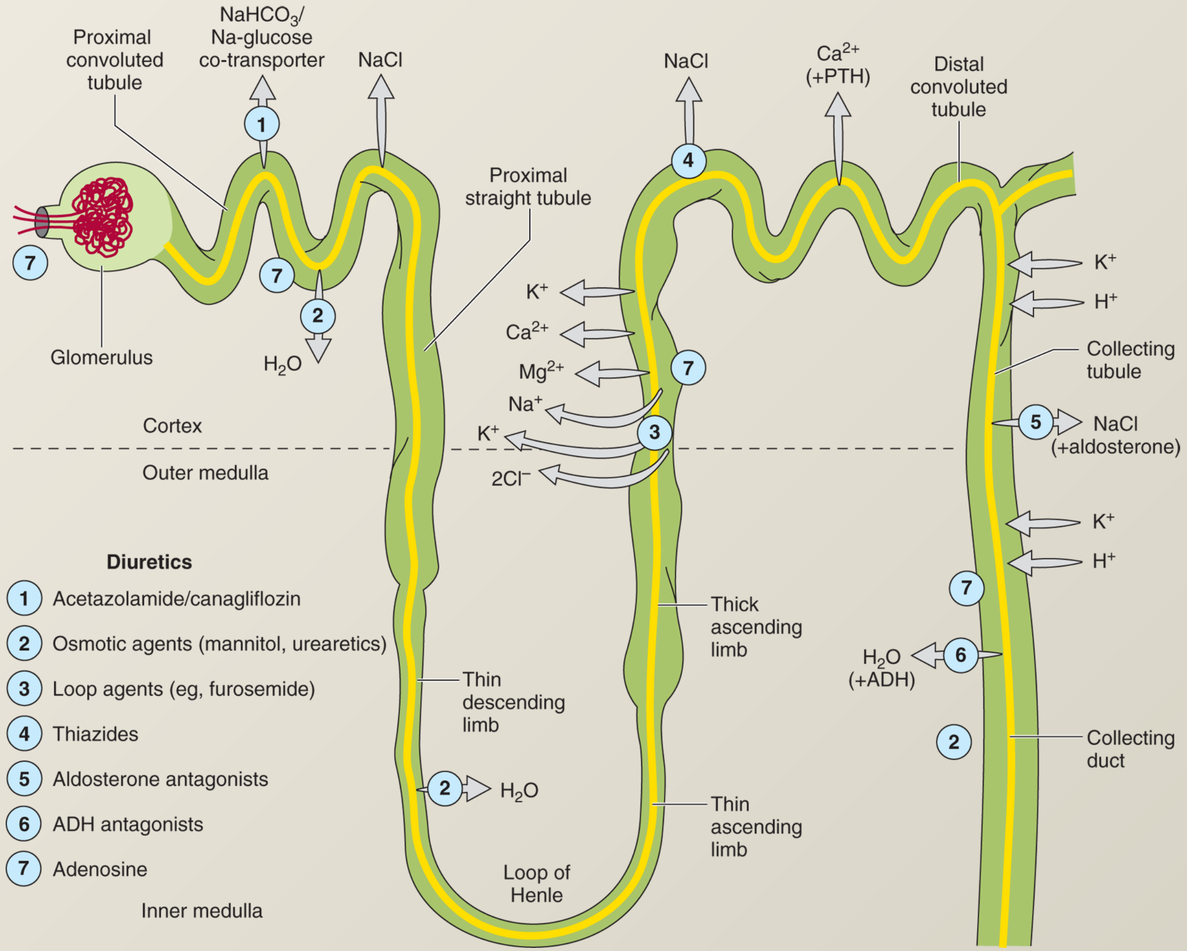
FIGURE 9.2 Effects of a loop diuretic on urinary sodium (Na^+) excretion.

National Kidney Foundation Primer

Explain the rationale behind combination therapy with loop diuretics and thiazides

Diuretic Combination Therapy: Loop Diuretics and Thiazides

- Combination therapy can be used to treat patients refractory to loop diuretics or who become refractory after a initial response
- Synergy gained by using two different sites of action
- Loop diuretic + thiazide
  - Salt reabsorption in either TAL or DCT can increase when the other is blocked due to tubuloglomerular feedback
  - **Blockade of both TAL and DCT greatly increases efficacy**
- **Highly increased risk of K^+-wasting**
  - Not recommended for routine outpatient use



"Basic and Clinical Pharmacology," 16e.

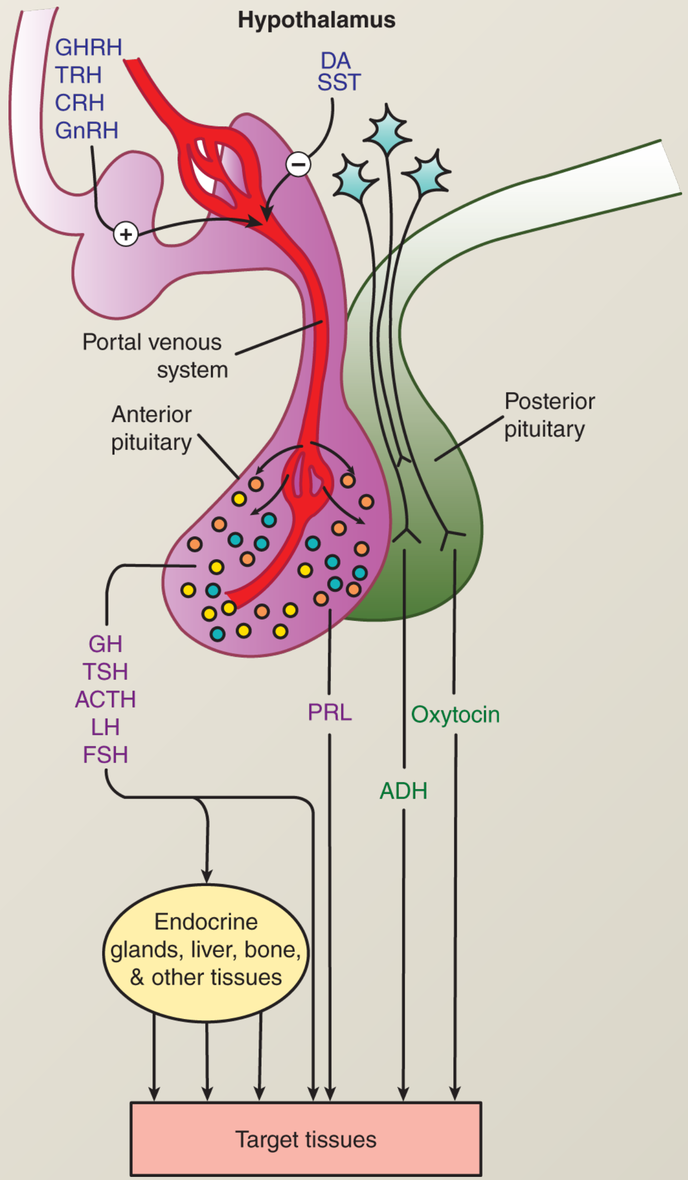
Explain the rationale behind combination therapy with loop diuretics and thiazides

Section 3: Vasopressin Agonists and Antagonists

- Objectives:
  - Review the physiology of vasopressin (ADH, AVP) release and actions at V1 and V2 receptors
  - Discuss the indications, pharmacodynamics, pharmacokinetics, and ADRs of vasopressin and desmopressin
  - Discuss the indications, pharmacodynamics, pharmacokinetics, and ADRs of vasopressin antagonists
  - Provide examples of pharmacokinetic and pharmacodynamic drug interactions with vasopressin receptor ligands

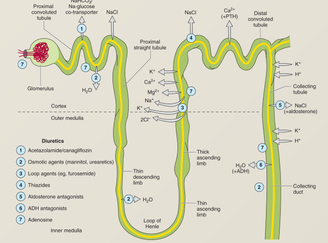
Physiology and regulation of vasopressin release

- Vasopressin (ADH or AVP) is released by the posterior pituitary gland
  - Hyperosmolality (> 290 mOsm/kg) stimulate release to act on target
  - Hypovolemia and hypotension
  - Neurotransmitters and pharmacological agents can alter ADH release
- Anterior pituitary hormones released after stimulation by hypothalamus



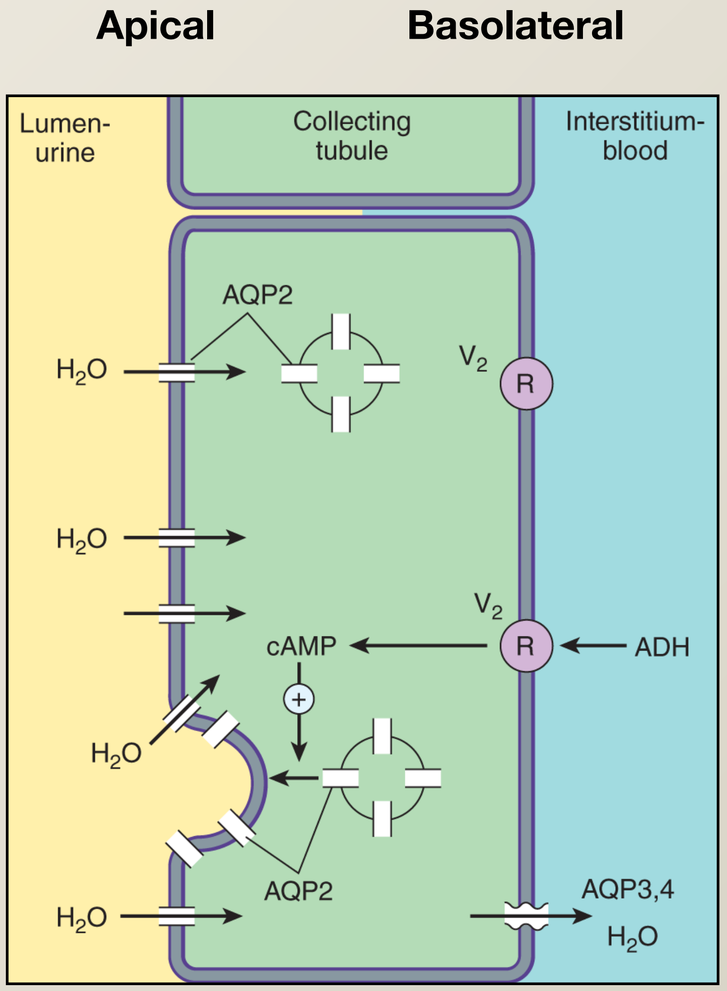
"Basic and Clinical Pharmacology," 16e.

Review the physiology of vasopressin (ADH, AVP) release and actions at V1 and V2 receptors



Vasopressin Agonists and Antagonists

- Agonists are used to treat diabetes insipidus!
- Antagonists are used to treat:
  - Syndrome of Inappropriate ADH Secretion (SIADH)
  - Other Causes of Elevated Antidiuretic Hormone (heart failure)
  - Autosomal Dominant Polycystic Kidney Disease

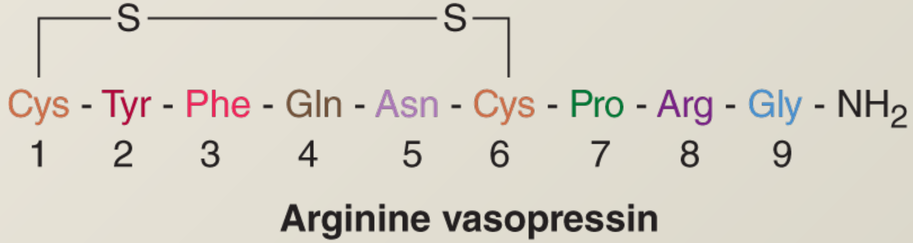


"Basic and Clinical Pharmacology," 16e.

Review the reabsorption mechanisms in the cortical collecting duct (CCD) and discuss the mechanism(s) of action, impact on electrolyte transport, ADRs, and indications of potassium-sparing diuretics

Vasopressin

- Vasopressin is a peptide hormone
  - a.k.a., antidiuretic hormone (ADH) or arginine vasopressin (AVP)
  - Disulfide bond between Cys residues forms 6-amino-acid ring
- Vasopressin acts at three subtypes of receptors:
  - V1aR, V1bR, and V2R

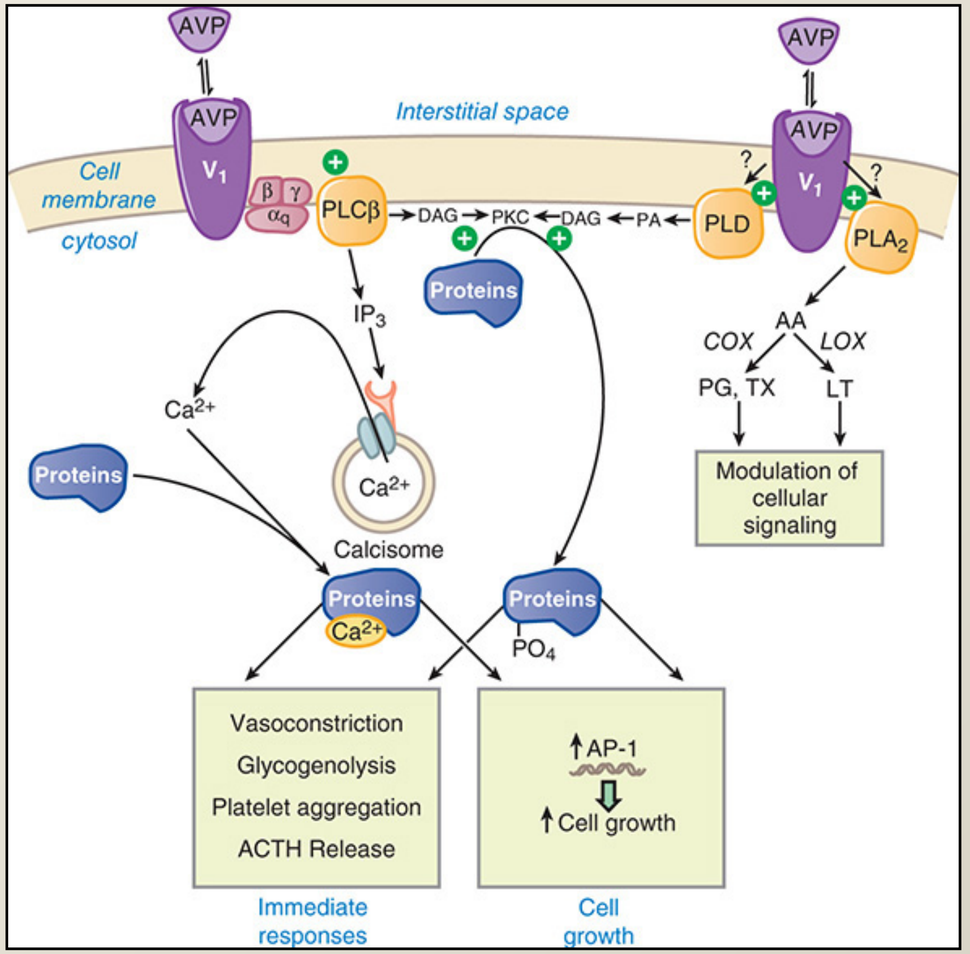


Arginine vasopressin

"Basic and Clinical Pharmacology," 16e.

Vasopressin receptors are GPCRs: V1R

- **V1R - Gα_q coupled**
  - Agonism increases phospholipase activity, intracellular Ca^{2+} release, and protein phosphorylation
- V1a receptors are **widely** expressed
  - vascular and enteric smooth muscle, myometrium (uterus)
  - adrenal gland, bladder, spleen, testis
  - adipocytes, hepatocytes, platelets
  - renal medulllary interstitium, vasa recta, renal cortical collecting duct epithelilal cell
- V1b receptors expressed in anterior pituitary, brain, pancreas, adrenal medulla

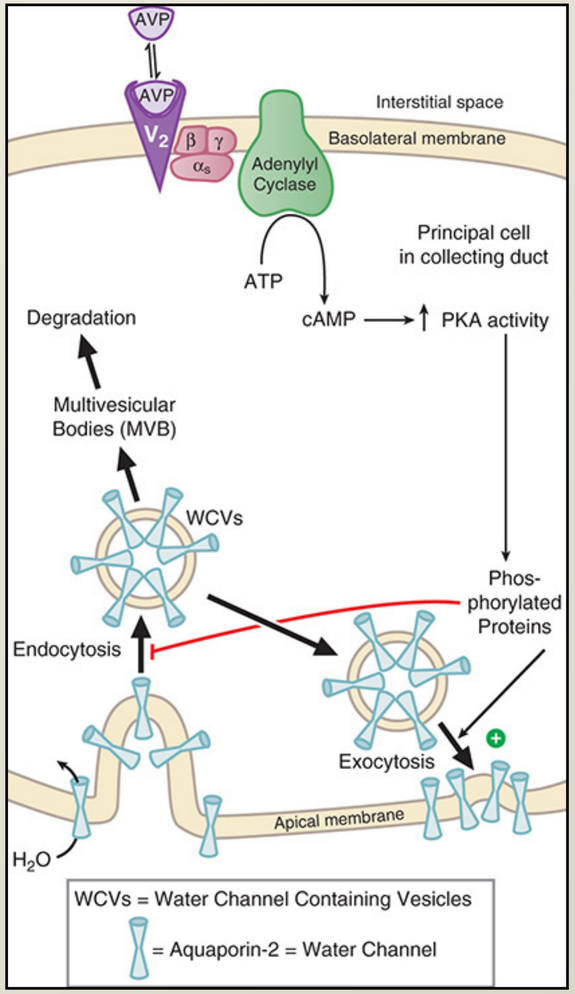


"Goodman & Gilman's The Pharmacological Basis of Therapeutics"

Review the physiology of vasopressin (ADH, AVP) release and actions at V1 and V2 receptors

Vasopressin receptors are GPCRs: V2R

- **V2R - Gα_s coupled**
  - Adenylyl cyclase is activated and increases cAMP production and PKA activity
- V2Rs are expressed in the collecting ducts
- Agonism of V2Rs expressed on principal cells increases trafficking of aquaporins to apical (lumen) membrane:
  - **Agonism increases water reabsorption**
  - **Antagonism increases diuresis**
- Activation of extrarenal V2's increases circulating levels of factor VIII and von Willebrand factor
  - Promotes blood coagulation

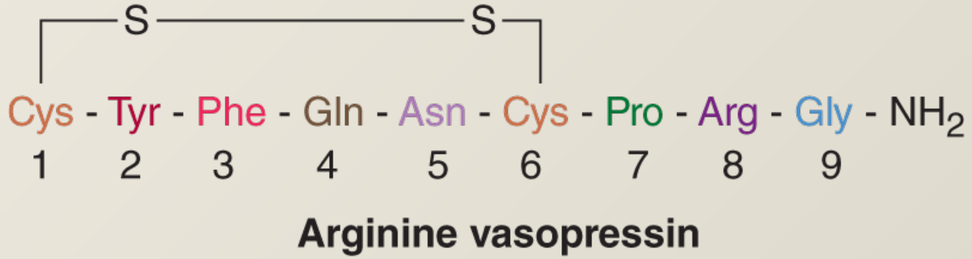


"Goodman & Gilman's The Pharmacological Basis of Therapeutics"

Review the physiology of vasopressin (ADH, AVP) release and actions at V1 and V2 receptors

Vasopressin Receptor Agonists

- *Vasopressin* (Vasostrict) is a non-selective agonist
  - Acts at **V1R** and **V2R** subtypes
    - Activity at V1R results in smooth muscle constriction
- *Desmopressin* (DDAVP) is a selective agonist
  - Acts at **V2R** subtype
    - Antidiuretic:pressor activity ratio is 3000-4000 times higher than for vasopressin
  - Long-acting synthetic analogue



Arginine vasopressin



Desmopressin

"Basic and Clinical Pharmacology," 16e.

DMPK of Vasopressin Agonists

- **Vasopressin** is available in aqueous solution
  - IV, IM, subcutaneous, intranasal
  - Inactivated by peptidases in liver and kidneys
  - $t_{1/2}$ ~ 15 minutes

- **Desmopressin** is available in aqueous solution and tablets
  - IV, IM, subcutaneous, intranasal
  - Nasal spray is available (4% bioavailability)
  - Inactivated by peptidases in liver and kidneys
    - Oral desmopressin inactivated by trypsin
  - $t_{1/2}$ ~ 75 minutes - 3.5 hours

Vasopressin Indications

- Indicated where GI and vascular smooth muscle **contraction** is beneficial:
  - Relief of postoperative ileus and abdominal distention
  - Reduction of bleeding in esophageal varices by constriction of portal vein
- Last-line agent for maintenance of arterial pressure in advanced cardiovascular life support
  - Shock due to trauma or sepsis
    - Used as adjuvant to reduce necessary catecholamine concentration
- May be used to treat central diabetes insipidus, not preferred

V1R-mediated effects	V2R-mediated effects
GI muscle contraction	Antidiuresis (Renal V2Rs)
Vascular muscle contraction	Blood coagulation (Extrarenal V2Rs)

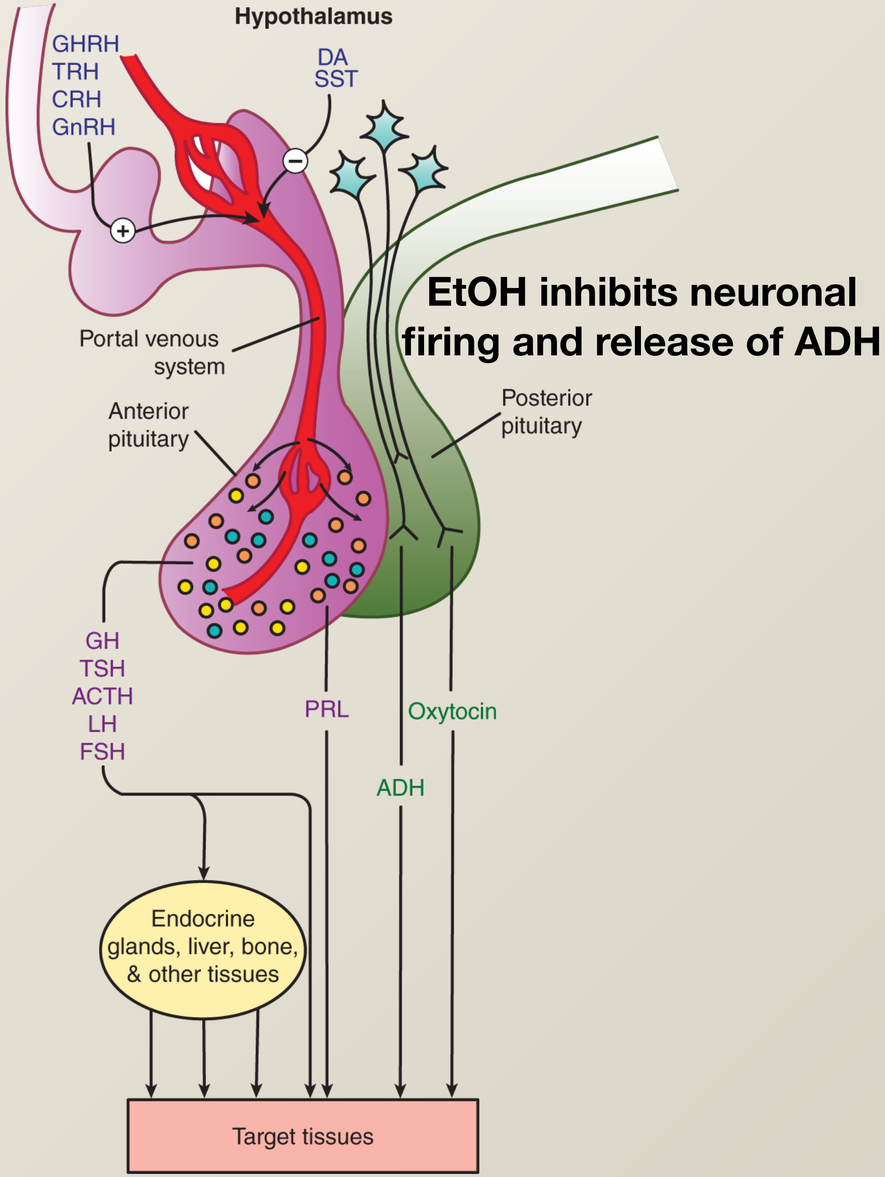
Vasopressin ADRs

- Vasoconstriction of coronary vessels
  - Angina, myocardial ischemia
- Doses required to observe vasoconstriction are usually exceed those needed for antidiuresis
- Contraindicated in **coronary insufficiency**
  - Deaths have resulted at therapeutic doses for Central DI
- Overdose can result in hyponatremia and seizures

V1R-mediated effects	V2R-mediated effects
GI muscle contraction	Antidiuresis (Renal V2Rs)
Vascular muscle contraction	Blood coagulation (Extrarenal V2Rs)

Ethanol and Vasopressin

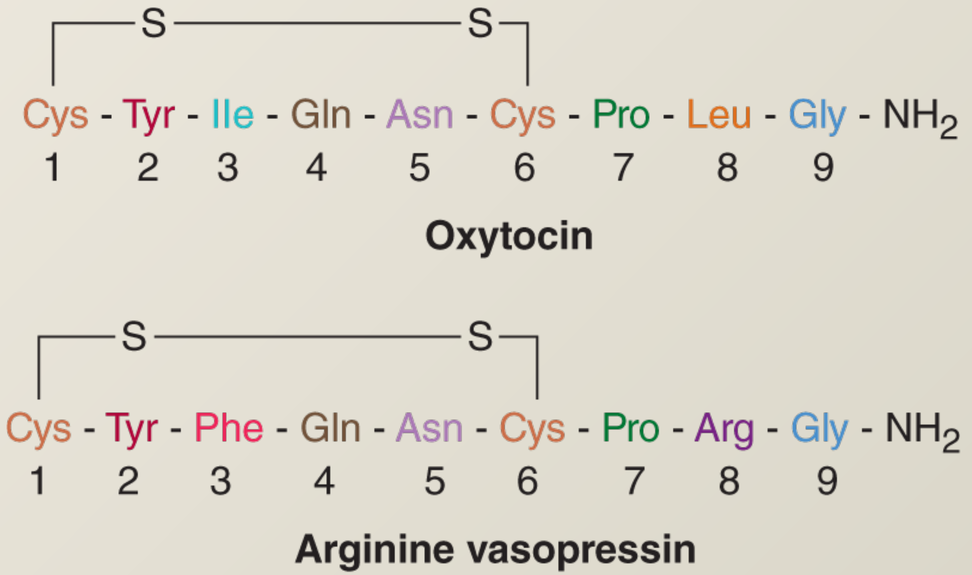
- Ethanol inhibits the release of vasopressin, resulting in enhanced diuresis
- Alcohol-dependent individuals in withdrawal exhibit increased vasopressin release
  - May experience water retention and hyponatremia



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Other nonrenal effects of Vasopressin

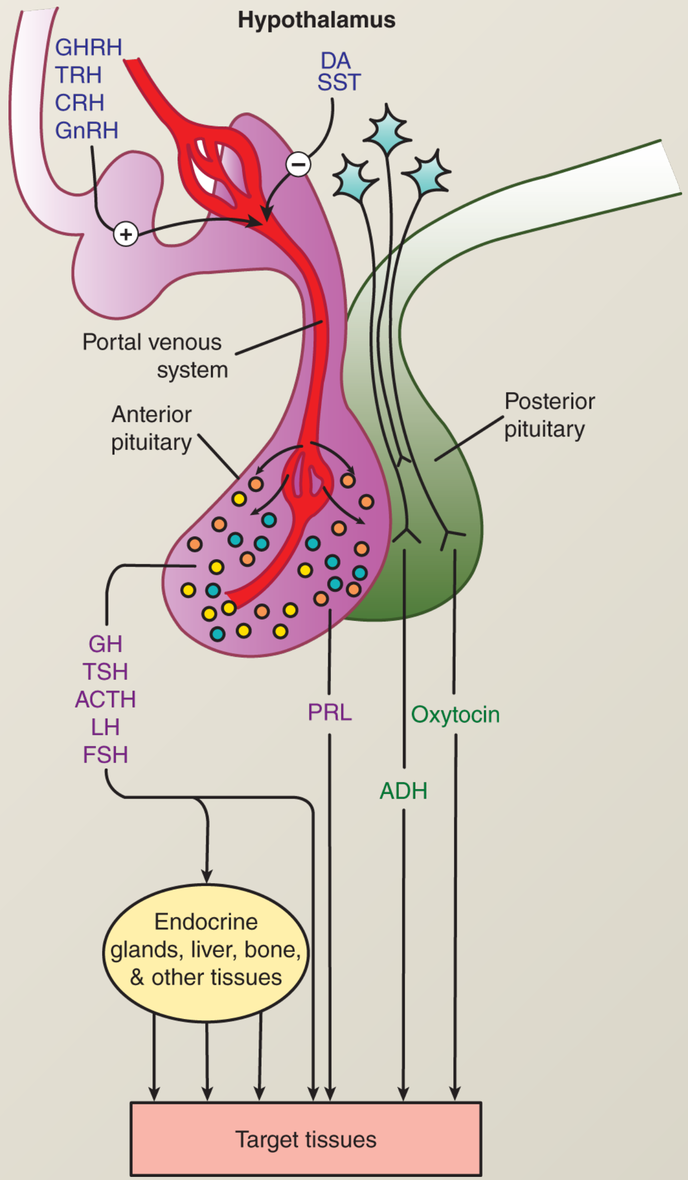
- Vasopressin has structural similarity to oxytocin
  - High concentrations of ADH stimulates smooth muscle contraction in uterus
  - May cause **premature labor**
- Agonism of V1 receptors on hepatocytes stimulates glycogenolysis
  - Increases blood glucose



"Basic and Clinical Pharmacology," 16e.

Other nonrenal effects of Vasopressin

- Can modulate autonomic signaling, but exact mechanisms not clearly established
- Activation of V1 receptors in anterior pituitary promote ACTH release
  - Vasopressin may promote sustained cortisol release in chronic stress
  - Reduced vasopressin levels may result in lower cortisol production



"Basic and Clinical Pharmacology," 16e.

Desmopressin Indications

- Often formulated as desmopressin acetate
- **Preferred treatment for central diabetes insipidus for many patients**
  - Intranasal or oral therapy is preferred
  - Approved for treatment of nocturnal enuresis
- Used in treatment of coagulopathies
  - Hemophilia A
  - von Willebrand disease

V1R-mediated effects	V2R-mediated effects
GI muscle contraction	Antidiuresis (Renal V2Rs)
Vascular muscle contraction	Blood coagulation (Extrarenal V2Rs)

Desmopressin ADRs

- Thrombosis occurs rarely, but may result in DVT, stroke, or myocardial infarction
- Overdose can result in hyponatremia and seizure
    - Patients treated for bleeding disorders at risk due to frequent fluid replacement

V1R-mediated effects	V2R-mediated effects
GI muscle contraction	Antidiuresis (Renal V2Rs)
Vascular muscle contraction	Blood coagulation (Extrarenal V2Rs)

Diabetes Insipidus

- **Central Diabetes Insipidus (Central DI):**
  - Inadequate production or secretion of ADH by the pituitary
    - May have hereditary, idiopathic, or injury causes
- **Diabetes insipidus** in pregnancy
  - Increased levels of circulating vasopressinase and decreased renal vasopressin sensitivity can unmask or accentuate preexisting DI
- **Nephrogenic Diabetes Insipidus (Nephrogenic DI):**
  - Inadequate response of the kidney to ADH
    - Hereditary (90% of cases due to mutations in V2)
    - Drug-induced
    - Electrolyte imbalance-induced

Central diabetes insipidus is treated with a vasopressin agonist (desmopressin)

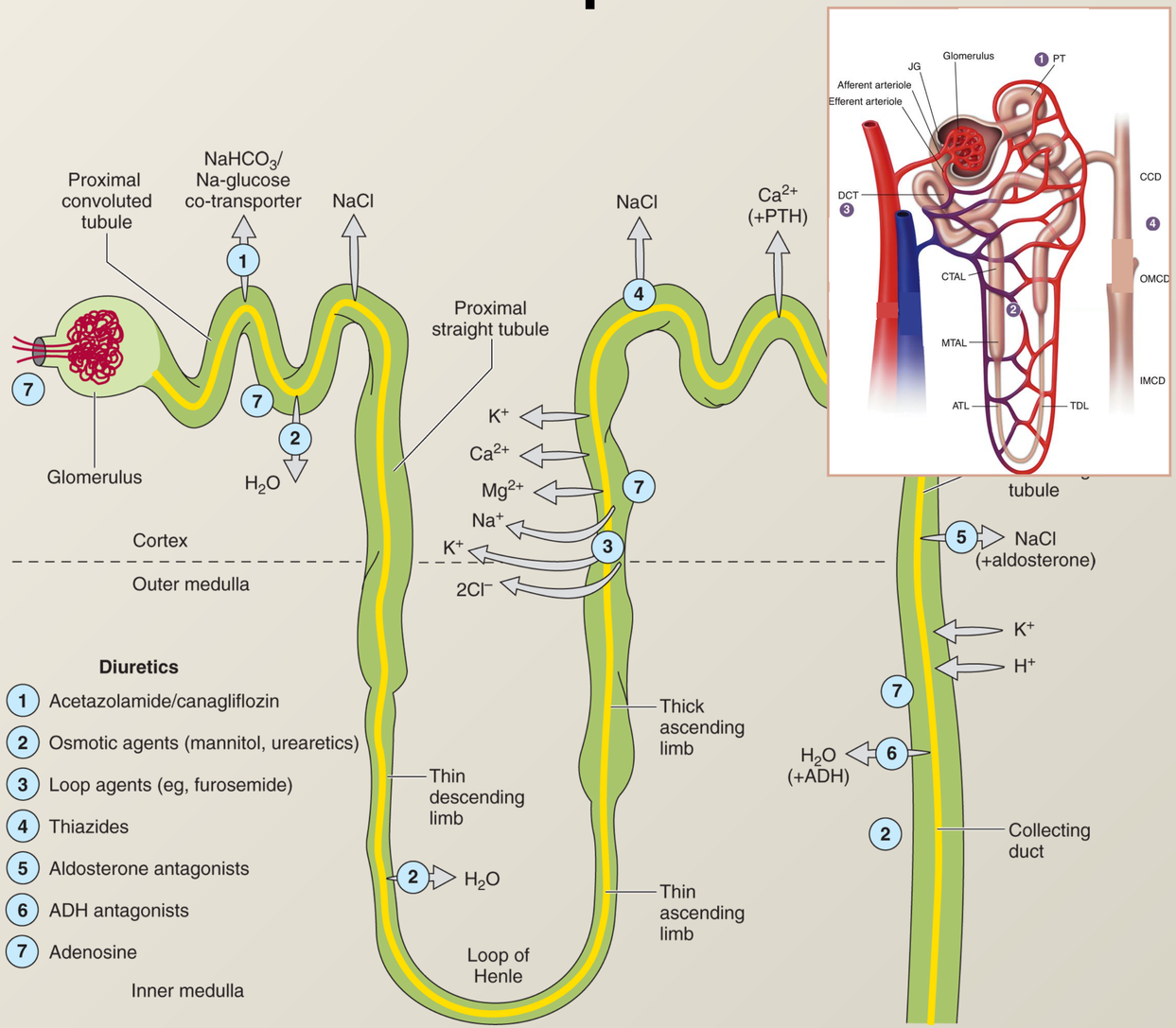
Gestational diabetes insipidus is treated with a vasopressin agonist (desmopressin)

Treatment of nephrogenic diabetes insipidus varies based on mechanism of origin

"Goodman & Gilman's The Pharmacological Basis of Therapeutics"

Treatment of Nephrogenic Diabetes Insipidus

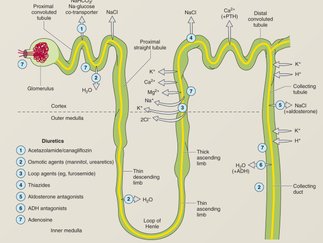
- Thiazide therapy is indicated for nephrogenic diabetes insipidus **when the defect is in collecting tubule/ duct**
- **Thiazide** treatment thought to produce *mild hypovolemia*, which reduces renal blood flow and GFR
  - Slower filtrate movement allows more reabsorption in proximal tubule
  - Thiazide actions in the TAL do not **directly** contribute to urine concentration
- Prostaglandin synthesis inhibitors (NSAIDs) accentuate thiazide action



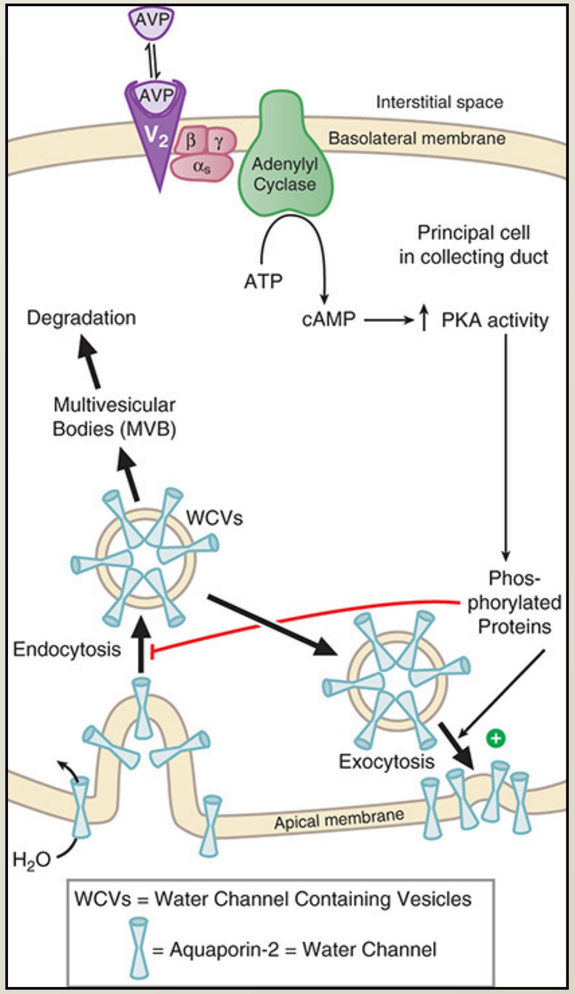
"Basic and Clinical Pharmacology," 16e.

Fig. 26-5, *Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy,* 5e

Lithium-induced Nephrogenic diabetes insipidus

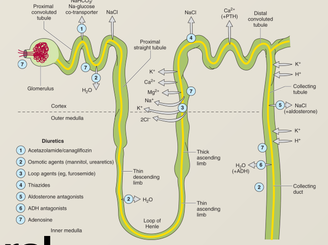


- **Li^+ is a common treatment for bipolar disorder**
- Li^+-induced NDI occurs in ~1 out of 3 patients
  - Li^+ reduces cAMP production stimulated by V2 receptor agonism
  - Increased plasma levels of PTH
- Demecocylcline (a tetracycline) and foscarnet (Foscavir) also inhibit cAMP production

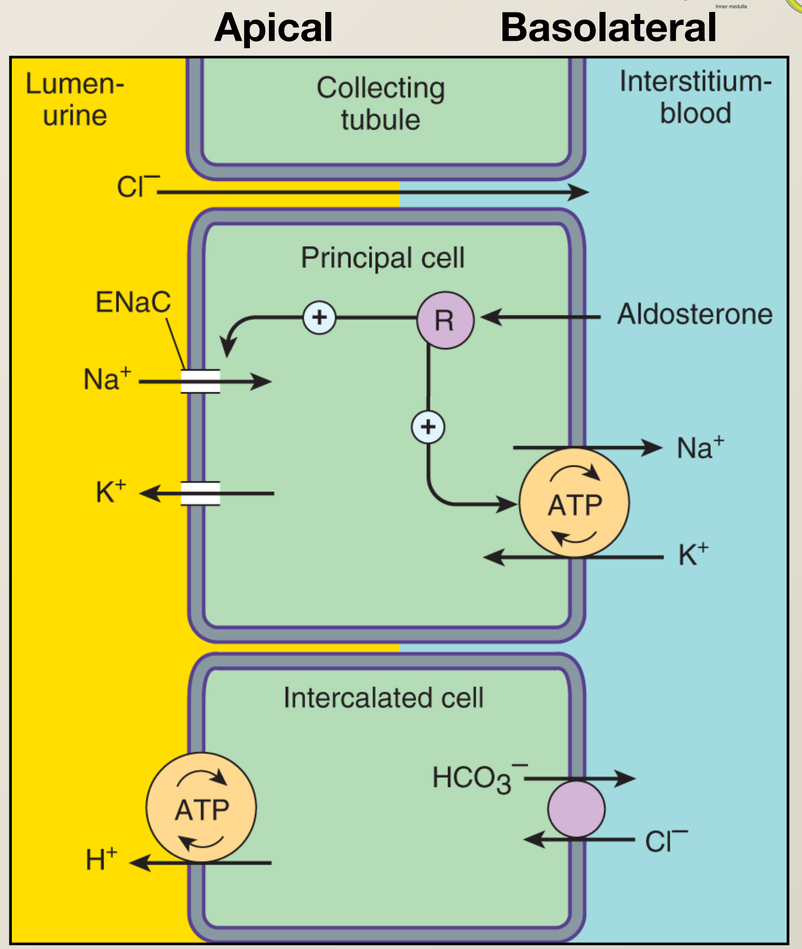


"Goodman & Gilman's The Pharmacological Basis of Therapeutics"
"Basic and Clinical Pharmacology," 14th Ed"Basic and Clinical Pharmacology," 16e.

Amiloride treats Lithium-induced NDI



- Li^+ is in the same chemical group as Na^+
  - Some Na^+ channels (ENaC) are permeable to Li^+
- Amiloride blocks ENaC, prevents Li^+ reabsorption from collecting tubule
  - Reduces impact on V2 signaling



"Basic and Clinical Pharmacology," 16e.

Vasopressin Antagonists

- ***Antagonists of vasopressin receptors promote diuresis***
- **Lithium** and **demeclocycline** inhibit V2R-mediated cAMP production
- **Conivaptan** (Vasiprol) targets both V1aR and V2R
- **Tolvaptan** (Samsca) selectively targets V2R

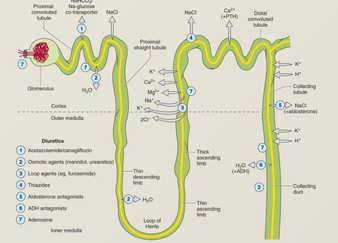
V1R-mediated effects	V2R-mediated effects
GI muscle contraction	Antidiuresis (Renal V2Rs)
Vascular muscle contraction	Blood coagulation (Extrarenal V2Rs)

Vasopressin antagonists will block these effects!

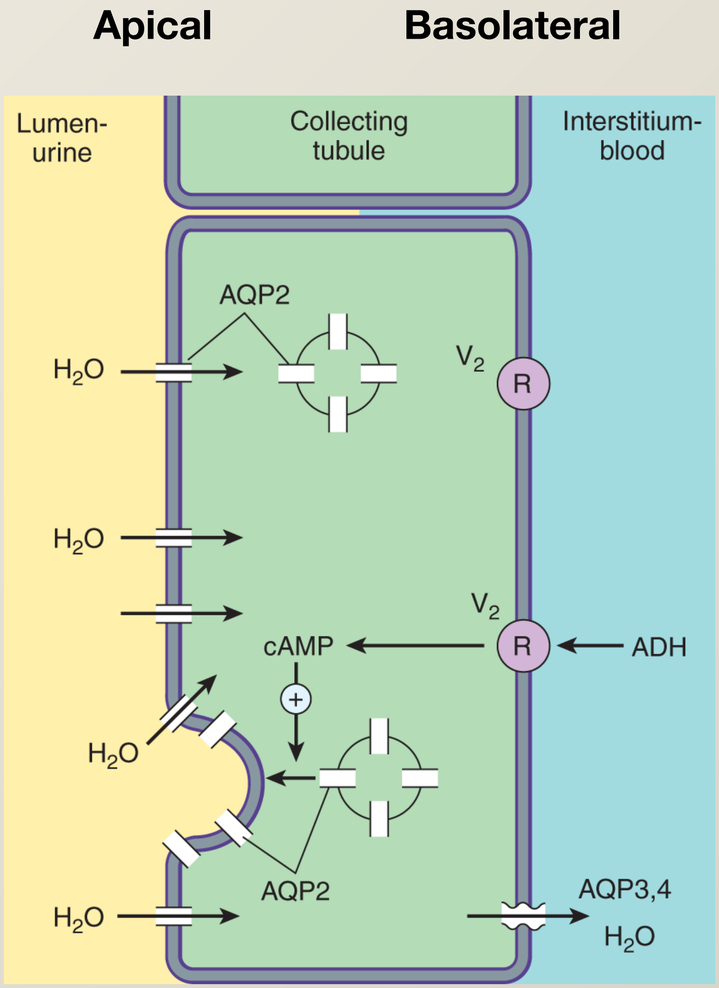
Vasopressin Antagonists: Vaptans

- **Conivaptan** is available for IV administration only
  - $t_{1/2}$ = 5-10 hours
  - Metabolized by CYP3A4
- **Tolvaptan** is available for oral administration
  - $t_{1/2}$ = 12-24 hours
  - Metabolized by CYP3A4

Indications of Vasopressin Antagonists



- Vaptans are used to treat volume overload conditions
  - ***Aquaretics* increase urine volume by excretion of solute-free water**
  - Vaptans decrease volume overload without affecting other solutes
- Indicated for conditions that result in **hyponatremia** (hypervolemic and euvolemic)
- Autosomal Dominant Polycystic Kidney Disease



"Basic and Clinical Pharmacology," 16e.

Indications of Vasopressin Antagonists: Syndrome of inappropriate ADH (SIADH) secretion

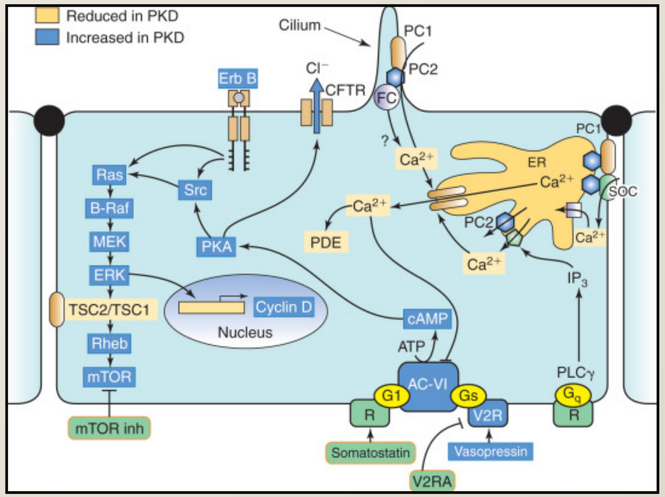
- Antidiuretic hormone antagonists are used to manage SIADH when water restriction has failed to correct the abnormality
- **Demeclocycline** or **tolvaptan**
  - Appropriate plasma levels of demeclocycline should be maintained by monitoring
  - Demeclocycline is not appropriate for patients under the age of 12

Indications of Vasopressin Antagonists: Heart Failure

- ADH is elevated with diminished effective circulating blood volume (e.g., heart failure)
  - Hyponatremia
- Water restriction is often treatment of choice (adherence issues)
- **Conivaptan** affinity for V1aR decreases peripheral vascular resistance and increases cardiac output
- **Tolvaptan** is used as an adjunct to diuretic therapy in CHF patients
- As many as 1 in 5 patients hospitalized for acute HF present with serum sodium concentrations less than 136 mmol/L
  - Associated with increased mortality

Indications of Vasopressin Antagonists: Autosomal Polycystic Kidney Disease

- Mutations in polycystin proteins result in changes in the intracellular calcium level or increases in the level of cAMP
- Altered intracellular signaling pathways (E.g., Ras, mTOR)
- Changes in apoptosis and fluid secretion occur
  - Thought to contribute to cyst development
- Blockade of V2R's with vaptans can reduce cAMP levels and delay cyst formation:
  - In a large multicenter prospective trial, **tolvaptan** reduced increase in kidney size and slowed progression of kidney failure over 3 years
  - Major ADR observed was liver toxicity



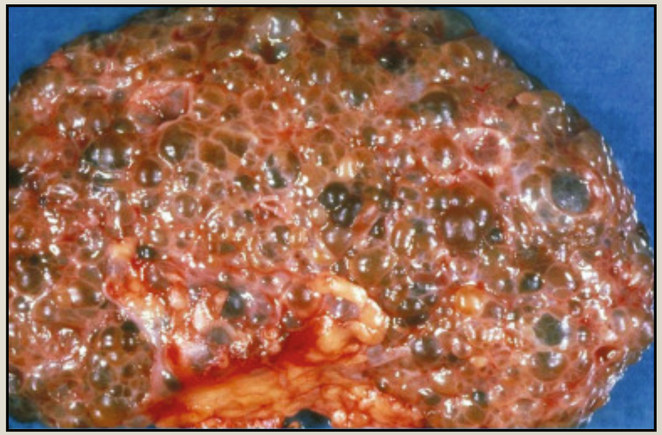


Fig. 40-1 and 40-2; National Kidney Foundation Primer

Vasopressin Antagonists: ADRs

- **Dry mouth** and **thirst** are common
- **Tolvaptan** may cause **hypotension**
  - Tolvaptan may also cause an **elevation** in liver function tests
  - Contraindicated in patients with liver disease
  - Drug use should be limited to 30 days
- **Lithium** and **demeclocylcine** may cause renal failure
  - Long-term lithium therapy may also cause chronic interstitial nephritis
  - Demeclocycline should be avoided in patients with liver disease and in children younger than 12 years
- **Nephrogenic Diabetes Insipidus**
  - Any ADH antagonist can cause severe hypernatremia and nephrogenic diabetes insipidus
    - Monitor serum sodium closely!

Conivaptan: ADRs

- N = 251 hyponatremic patients
- Most common ADR reported (48.2% of patients) was phlebitis
- GI disorders (nausea, vomiting, diarrhea, etc.) reported in 35%
- 8 patients were removed from the study due to ≥ 12 mEq/L increase in serum Na^+ in under 24 hours
  - Risk of osmotic demyelination syndrome

Drug Des Devel Ther. 2016; 10: 339–351. PMCID: PMC4723016
Published online 2016 Jan 18. doi: 10.2147/DDDT.S95326 PMID: 26848258

Dose comparison of conivaptan (Vaprisol®) in patients with euvolemic or hypervolemic hyponatremia – efficacy, safety, and pharmacokinetics

Biff F Palmer,[1] Amy D Rock,[2] and Emily J Woodward[2]

Osmotic demyelination syndrome

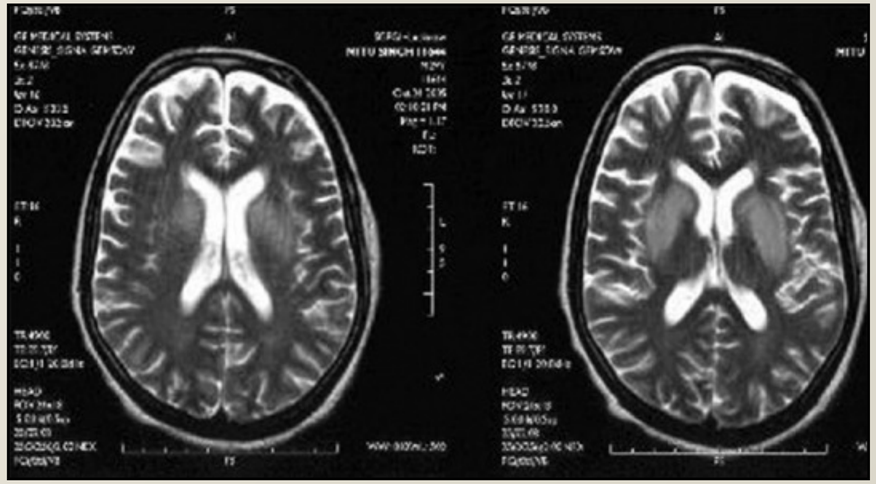
- Occurs due to rapid rise in serum sodium levels in hyponatremic patients
- Rapid fluctuations in ECF/ICF damage myelin sheath
  - Rare, but alcoholism and SIADH raise risk significantly
- Tolvaptan has black box warning

Indian J Crit Care Med. 2015 Mar; 19(3): 166–169.
PMCID: PMC4366916
doi: 10.4103/0972-5229.152760
PMID: 25810613

Osmotic demyelination syndrome in Intensive Care Unit

Parnandi Bhaskar Rao, Afzal Azim,[1] Neha Singh,[2] Arvind Kumar Baronia,[1] Anand Kumar,[1] and Banani Poddar[1]

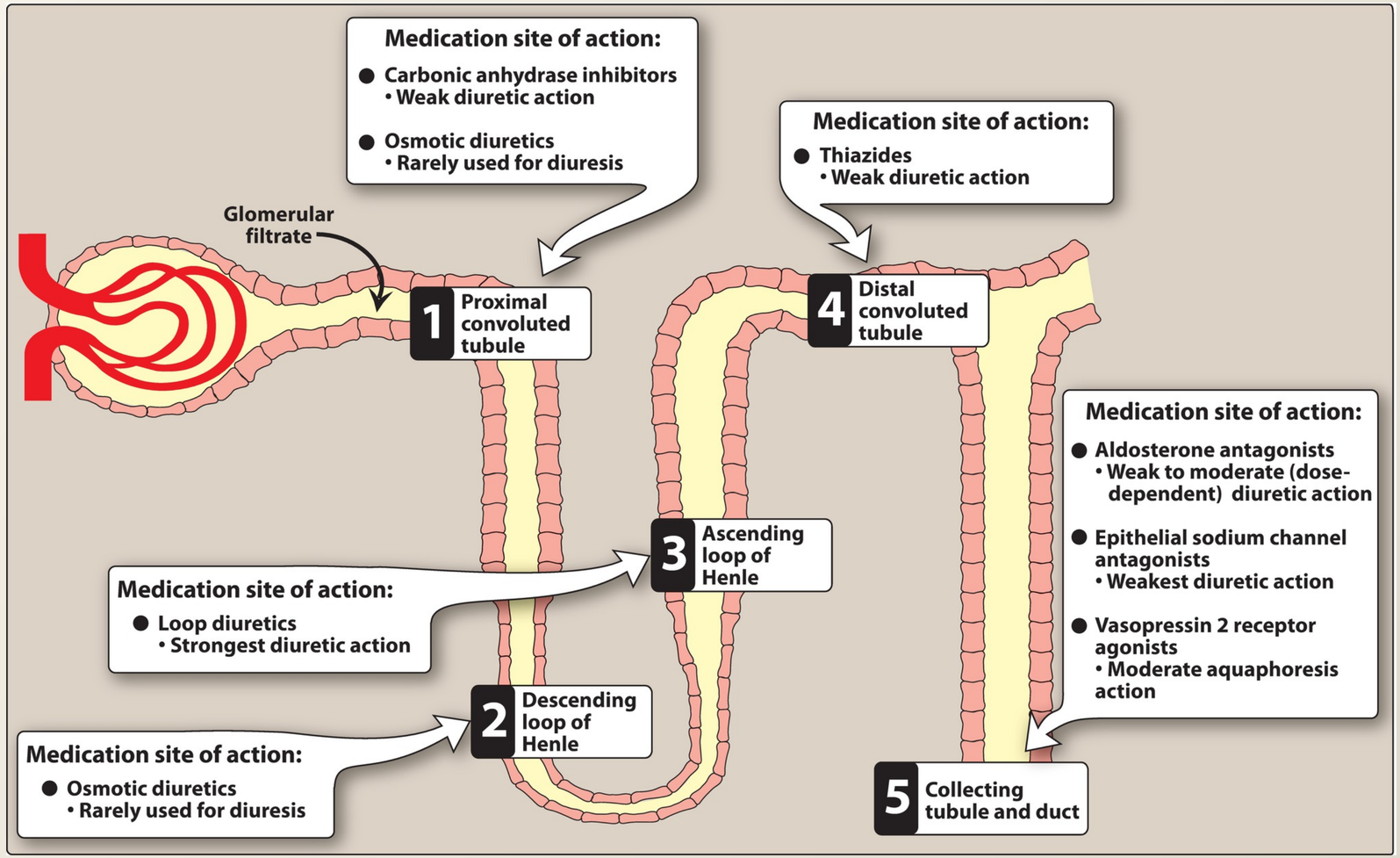
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T2-weighted magnetic resonance imaging showing areas of demyelination in bilateral basal ganglia and thalami, known as extra-pontine myelinolysis

Summary and Review Resources

Summary and Relative Potencies of Diuretics



Lippincott Illustrated Reviews, 8e

Summary slide of pH effects and urine electrolyte levels

DIURETIC CLASS	URINE VOLUME	URINARY EXCRETION OF:						
		Na^+	K^+	Mg^{2+}	Ca^{2+}	Cl^-	HCO_3^-	Uric acid
Thiazide	Initial: ↑ Chronic: ↔	↑	↑	↑	↓	↑	↓	↓
Loop	↑↑↑	↑↑	↑↑	↑	↑↑	↑	↓↓	↓
Potassium sparing								
Aldosterone antagonists	↑	↔	↓	↔	↔	↔	↔	↔
Epithelium sodium channel antagonists	↔	↔	↓	↔	↔	↔	↔	↔
Carbonic anhydrase inhibitor	↑	↔	↑	↔	↔	↔	↑	↔

Lippincott Illustrated Reviews, 8e

TABLE 15–2

Changes in urinary electrolyte patterns and body pH in response to diuretic drugs.

Group	Urinary Electrolytes			Body pH
	NaCl	$NaHCO_3$	K^+	
Carbonic anhydrase inhibitors	+	+++	+	↓
Loop agents	++++	0	+	↑
Thiazides	++	+	+	↑
Loop agents plus thiazides	+++++	+	++	↑
K^+-sparing agents	+	(+)	−	↓

+, increase; −, decrease; 0, no change; ↓, acidosis; ↑, alkalosis.

“Basic and Clinical Pharmacology,” 16e.

Summary Slide

LOOP DIURETICS				
• Furosemide	Inhibition of the Na/K/2Cl transporter in the ascending limb of Henle's loop	Marked increase in NaCl excretion, some K wasting, hypokalemic metabolic alkalosis, increased urine Ca and Mg	Pulmonary edema, peripheral edema, heart failure, hypertension, acute hypercalcemia, anion overdose	Oral and parenteral preparations • duration of action 2–4 h • *Toxicity:* Ototoxicity, hypovolemia, K wasting, hyperuricemia, hypomagnesemia
• *Bumetanide, torsemide: Sulfonamide loop agents like furosemide*				
• *Ethacrynic acid: Not a sulfonamide but has typical loop activity and some uricosuric action*				
THIAZIDES				
• Hydrochlorothiazide	Inhibition of the Na/Cl transporter in the distal convoluted tubule	Modest increase in NaCl excretion • some K wasting • hypokalemic metabolic alkalosis • decreased urine Ca	Hypertension, mild heart failure, nephrolithiasis, nephrogenic diabetes insipidus	Oral • duration 8–12 h • *Toxicity:* Hypokalemic metabolic alkalosis, hyperuricemia, hyperglycemia, hyponatremia
• *Metolazone: Popular for use with loop agents for synergistic effects*				
• *Chlorothiazide: Only parenteral thiazide available (IV)*				
• *Chlorthalidone: Long half-life (50–60 h) due to binding to red blood cells*				

"Basic and Clinical Pharmacology," 14th Ed.

Summary Slide

POTASSIUM-SPARING DIURETICS				
• Spironolactone	Pharmacologic antagonist of aldosterone in collecting tubules • weak antagonism of androgen receptors	Reduces Na retention and K wasting in kidney • poorly understood antagonism of aldosterone in heart and vessels	Aldosteronism from any cause • hypokalemia due to other diuretics • postmyocardial infarction	Slow onset and offset of effect • duration 24–48 h • *Toxicity:* Hyperkalemia, gynecomastia (spironolactone, not eplerenone) • additive interaction with other K-retaining drugs
• Amiloride	Blocks epithelial sodium channels in collecting tubules	Reduces Na retention and K wasting • increases lithium clearance	Hypokalemia from other diuretics • reduces lithium-induced polyuria • Liddle's syndrome	Orally active • duration 24 h • *Toxicity:* Hyperkalemic metabolic acidosis
• *Eplerenone: Like spironolactone, more selective for aldosterone receptor*				
• *Triamterene: Mechanism like amiloride, much less potent, more toxic*				
OSMOTIC DIURETICS				
• Mannitol	Physical osmotic effect on tissue water distribution because it is retained in the vascular compartment	Marked increase in urine flow, reduced brain volume, decreased intraocular pressure, initial hyponatremia, then hypernatremia	Renal failure due to increased solute load (rhabdomyolysis, chemotherapy), increased intracranial pressure, glaucoma	IV administration • *Toxicity:* Nausea, vomiting, headache
VASOPRESSIN (ADH) ANTAGONISTS				
• Conivaptan	Antagonist at V_{1a} and V_2 ADH receptors	Reduces water reabsorption, increases plasma Na concentration, vasodilation	Hyponatremia, congestive heart failure	IV only, usually continuous • *Toxicity:* Infusion site reactions, thirst, polyuria, hypernatremia
• Tolvaptan	Selective antagonist at V_2 ADH receptors	Reduces water reabsorption, increases plasma Na concentration	Hyponatremia, SIADH	Oral • duration 12–24 h • *Toxicity:* Polyuria (frequency), thirst, hypernatremia

"Basic and Clinical Pharmacology," 14th Ed.

Quick Physiology Recall

Which segment of the nephron is responsible for the majority of sodium reabsorption?

A. Proximal Tubule
B. Loop of Henle (Ascending Limb)
C. Loop of Henle (Descending Limb)
D. Distal Convoluted Tubule
E. Collecting Duct

Which class(es) of agents do you think will produce the most profound diuresis? Why?

	Diuretic Agents	Site of Action
A	**Carbonic anhydrase inhibitors** (e.g., acetazolamide)	Proximal Tubule
B	**Osmotic diuretics** (e.g., mannitol)	Proximal Tubule; Loop of Henle (descending limb)
C	**Loop diuretics** (e.g., furosemide)	Loop of Henle (ascending limb)
D	**Thiazide and thiazide-like diuretics** (e.g., chlorothiazide)	Distal Convoluted Tubule
E	**Potassium-sparing diuretics** (e.g., amiloride, spironolactone)	Collecting Tubule/Duct
F	**ADH antagonists** (e.g., lithium, vaptan agents)	Collecting Duct

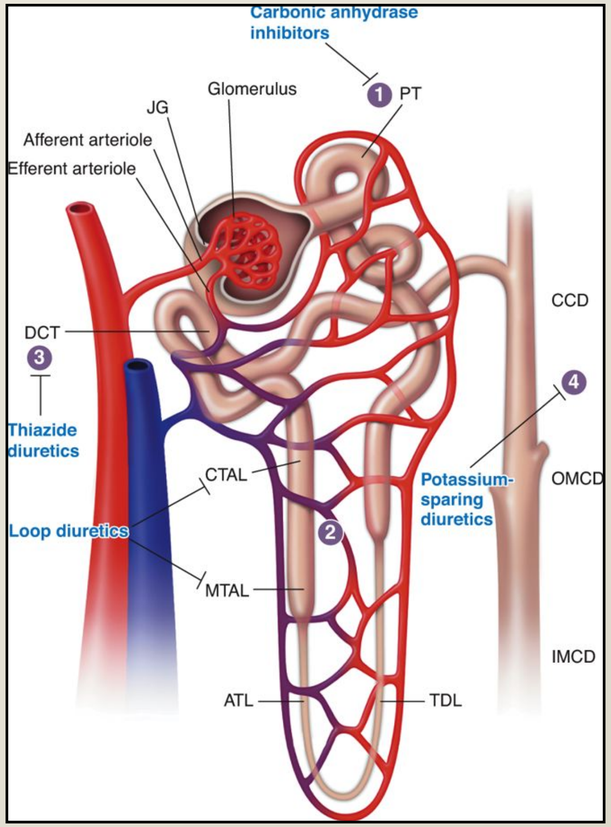


Fig. 21-5, *Principles of Pharmacology: The Pathophysiologic Basis of Drug Therapy*

Case

A 57-year-old male with acute angle-closure glaucoma is being treated with a diuretic to reduce intraocular pressure. Which of the following toxicities is most likely to occur in response to this drug?

A. Hypernatremia
B. Increased risk of UTI
C. Metabolic acidosis
D. Metabolic alkalosis
E. Ototoxicity

Case

A 64-year-old female with acute pulmonary edema is given an IV infusion of furosemide. The patient's serum potassium drops by 0.79 mEq/L. Which region(s) of the nephron tubule are the most likely the source of potassium wasting?

A. Proximal Tubule
B. Descending Limb
C. Thick Ascending Limb
D. Distal Convoluted Tubule
E. Collecting Tubule

Case

A 74-year-old female with a history of osteoporosis is being treated with a thiazide diuretic. Which of the following best explains why this drug regimen was chosen?

A. Thiazides do not induce metabolic acidosis
B. Thiazides alkalinize urine, preventing Ca^{2+} wasting
C. Thiazides disrupt the transepithelial voltage gradient that promotes Ca^{2+} excretion
D. Thiazides reduce the presentation of Na^{+} to collecting tubule, preventing K^{+} excretion

Case

A 56-year-old male patient with hepatic cirrhosis develops edema of the lower extremities, concomitant with abdominal swelling. Which of the following classes of diuretics is most appropriate to treat this patient?

A. Carbonic Anhydrase Inhibitors
B. Osmotic diuretics
C. Loop diuretics
D. Thiazides
E. K^+ sparing diuretics

Case

A 26-year-old female diagnosed with bipolar disorder is taking lithium. After a few weeks of adhering to her medication, she presents with polydipsia, polyuria, and nocturia. She reports fatigue and muscle cramps, and her urine is colorless. The patient is hypernatremic.

Which of the following drugs is most appropriate for use in this patient?

A. Amiloride
B. Desmopressin
C. Hydrochlorothiazide
D. Tolvaptan
E. Vasopressin

Case

A patient with chronic hypertension has just added hydrochlorothiazide to his regimen of furosemide. He presents to his primary care physician with headache and nausea, and the patient's lab values reveal a serum sodium of 133 mEq/L (normal range: 136 mEq/L - 145 mEq/L).

The patient is worried he will not be able to manage hyponatremia with fluid intake restriction, so the patient is given a prescription for a vasopressin antagonist to help him excrete the excess water and is sent home. In the next 24 hours, the patient presents to the emergency department with seizures, and reports having a headache. Serum sodium levels are 145 mEq/L.

What happened?

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What happened?